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(54) **SYSTEM AND TECHNIQUE FOR
METATARSAL REALIGNMENT WITH
REDUCED INCISION LENGTH**

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(57) **ABSTRACT**

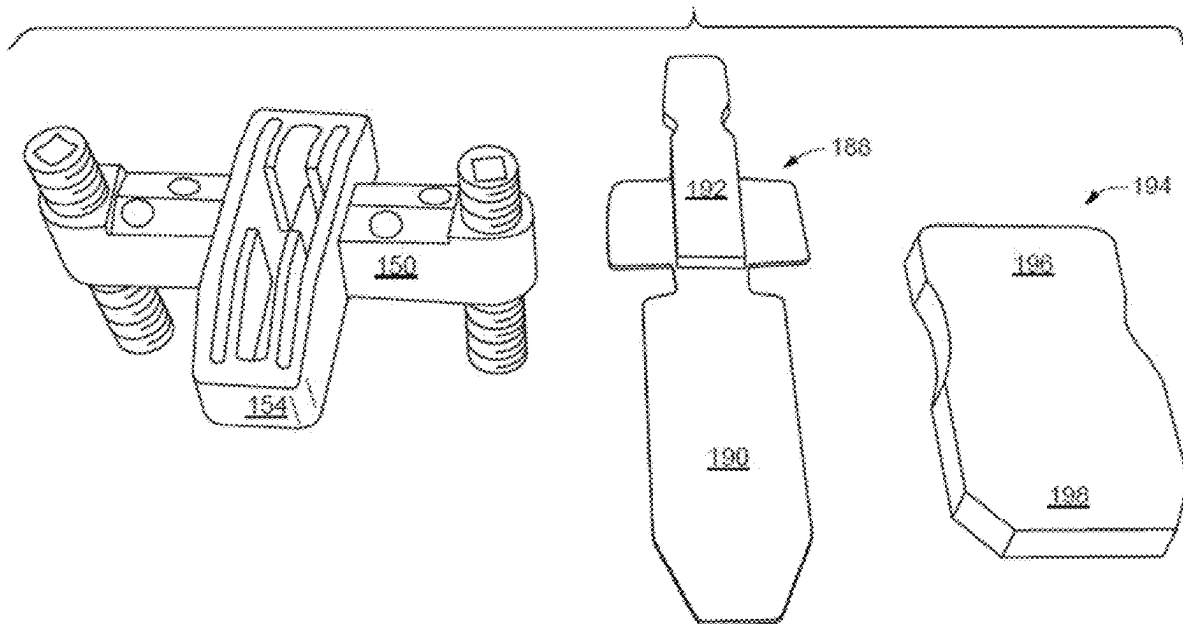
A metatarsal correction procedure may be performed so as to minimize the length of the surgical incision created during the procedure. A variety of different instruments and/or techniques and facilitate the minimal incision procedure. In some examples, a technique involves making a comparatively small surgical incision to access a tarsometatarsal joint. A metatarsal and a cuneiform separated by the tarsometatarsal joint can be prepared. In addition, the metatarsal can be moved relative to the cuneiform using a bone positioning device having a metatarsal engagement member positioned in contact with the skin of the patient on a medial side of the metatarsal. After moving the metatarsal, a fixation device can be installed across the tarsometatarsal joint to cause fusion. In some examples, a plate holder is attached to a U-shaped bone plate to facilitate plating across the joint through the comparatively small incision.

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Related U.S. Application Data

(63) Continuation of application No. 17/674,447, filed on Feb. 17, 2022, now Pat. No. 12,310,603.



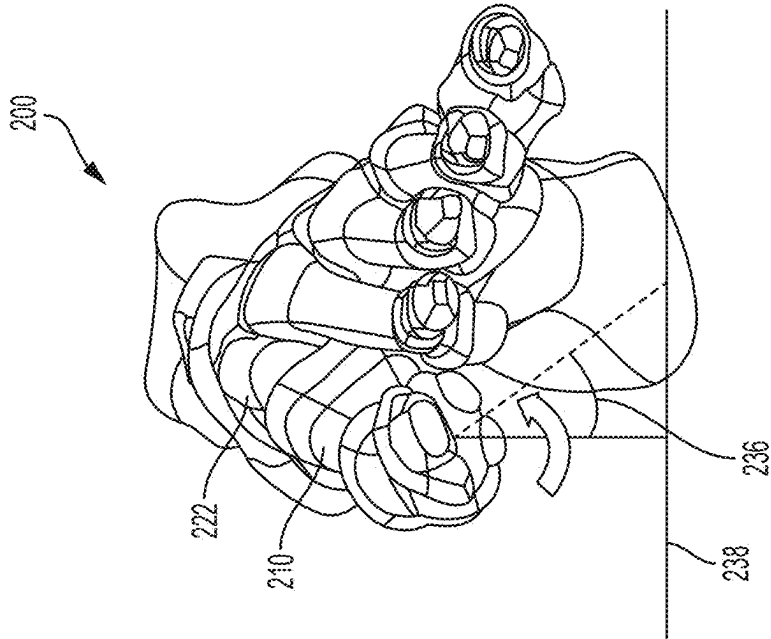


FIG. 1B

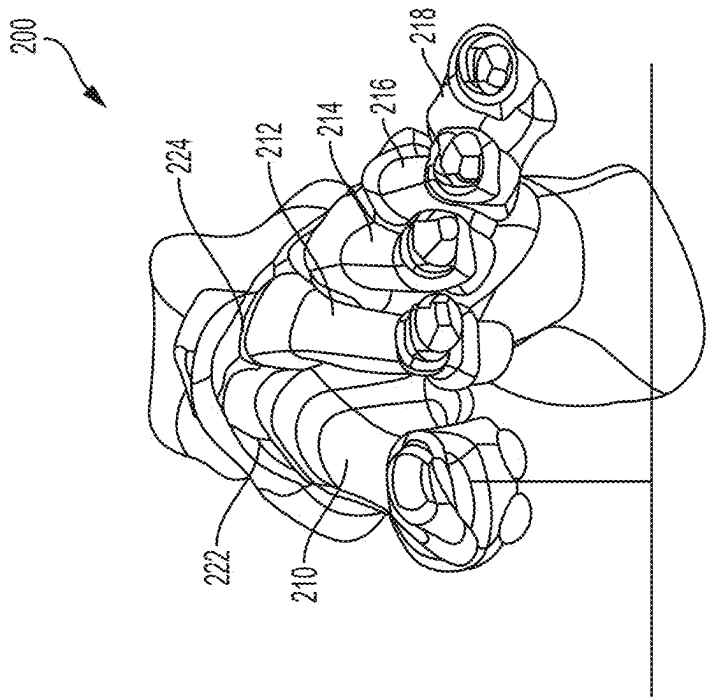


FIG. 1A

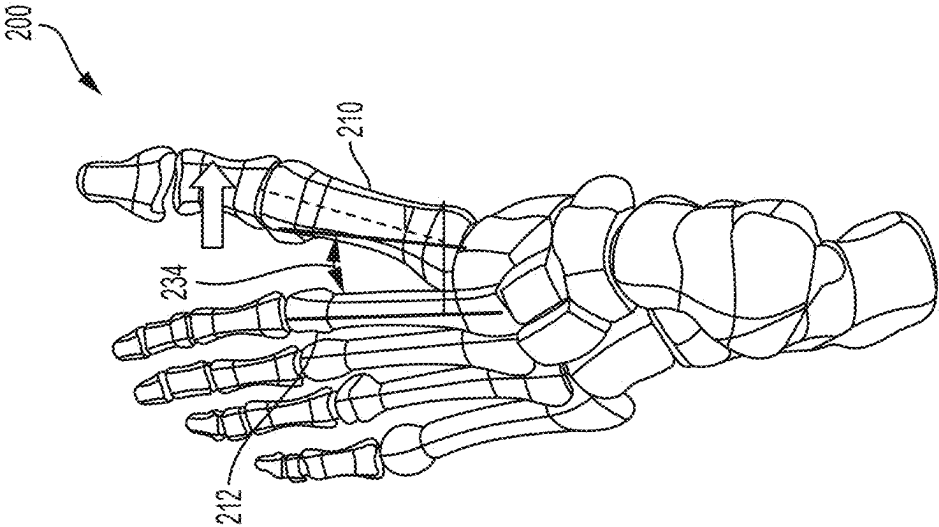


FIG. 2B

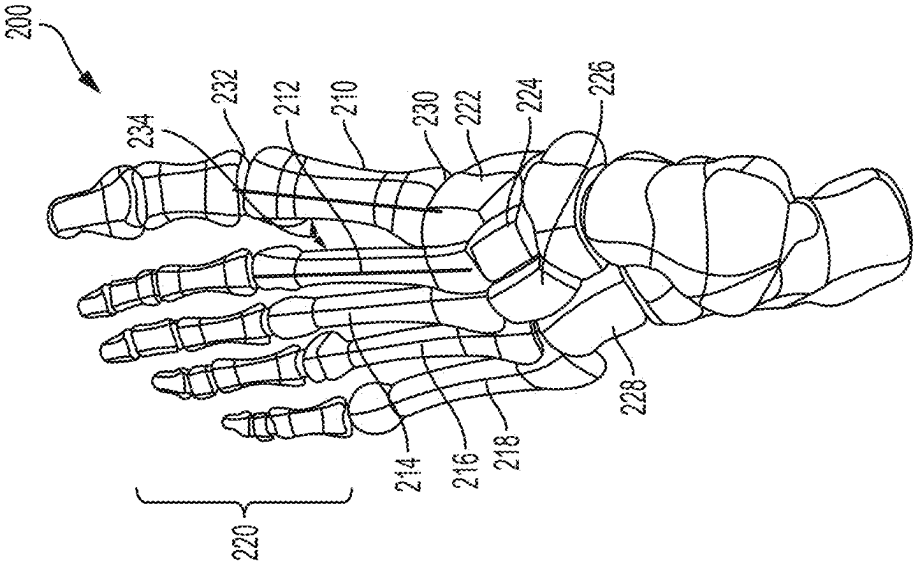


FIG. 2A

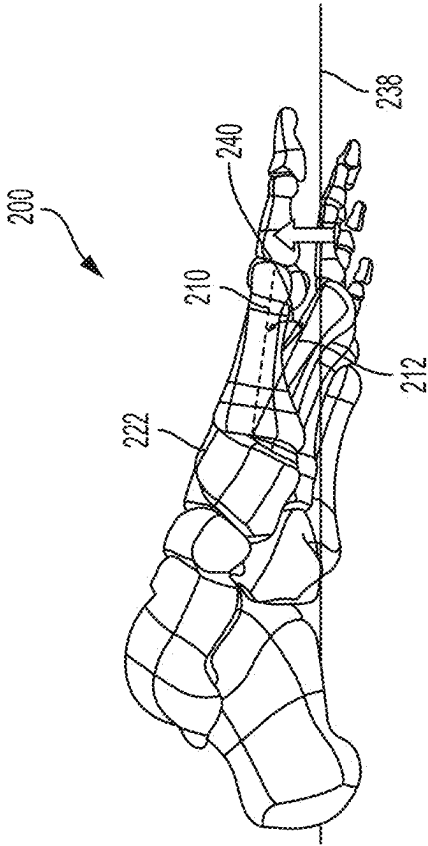


FIG. 3A

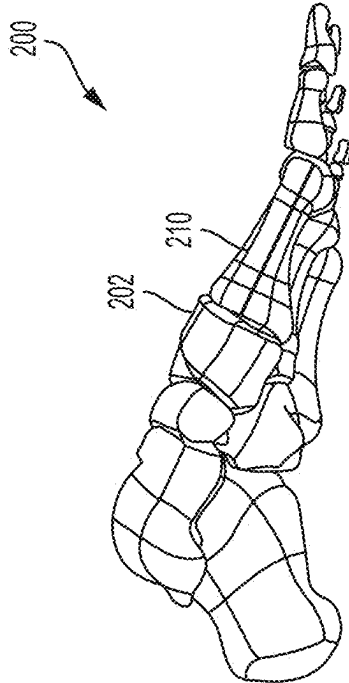


FIG. 3B

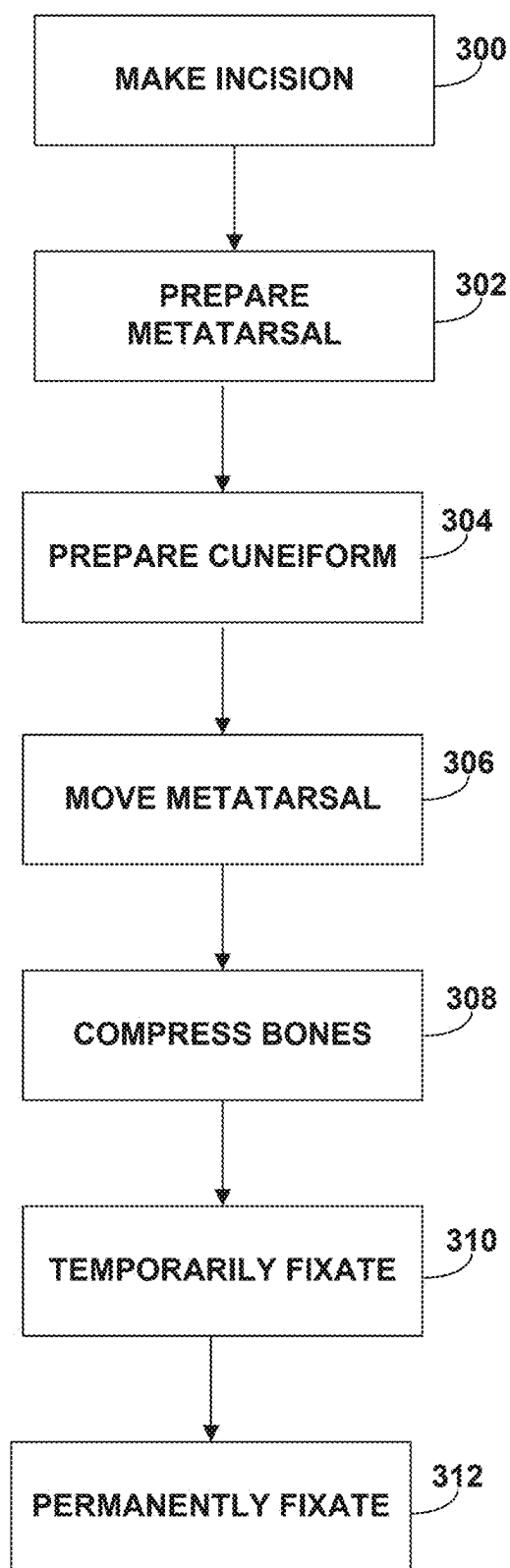


FIG. 4

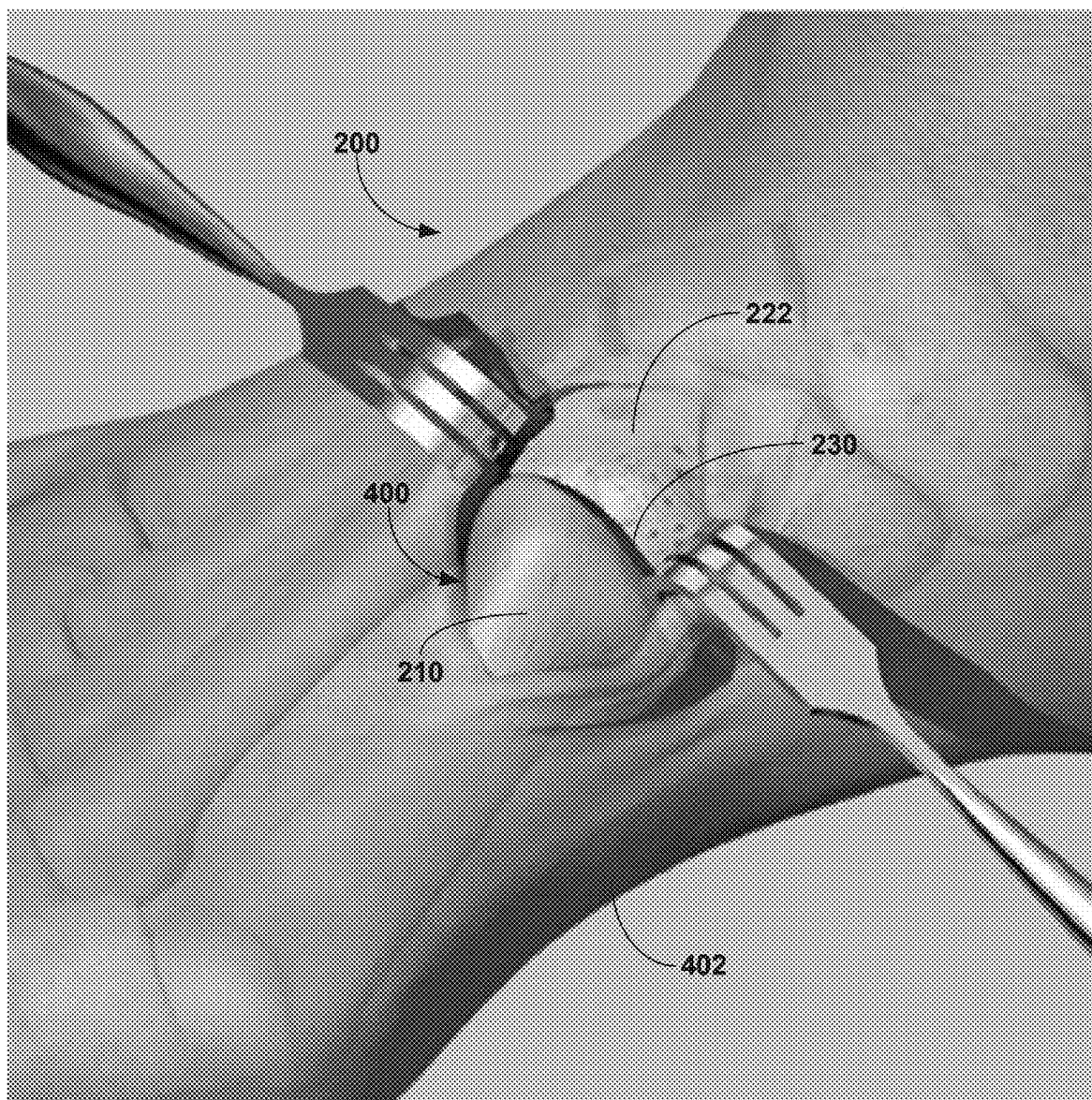


FIG. 5

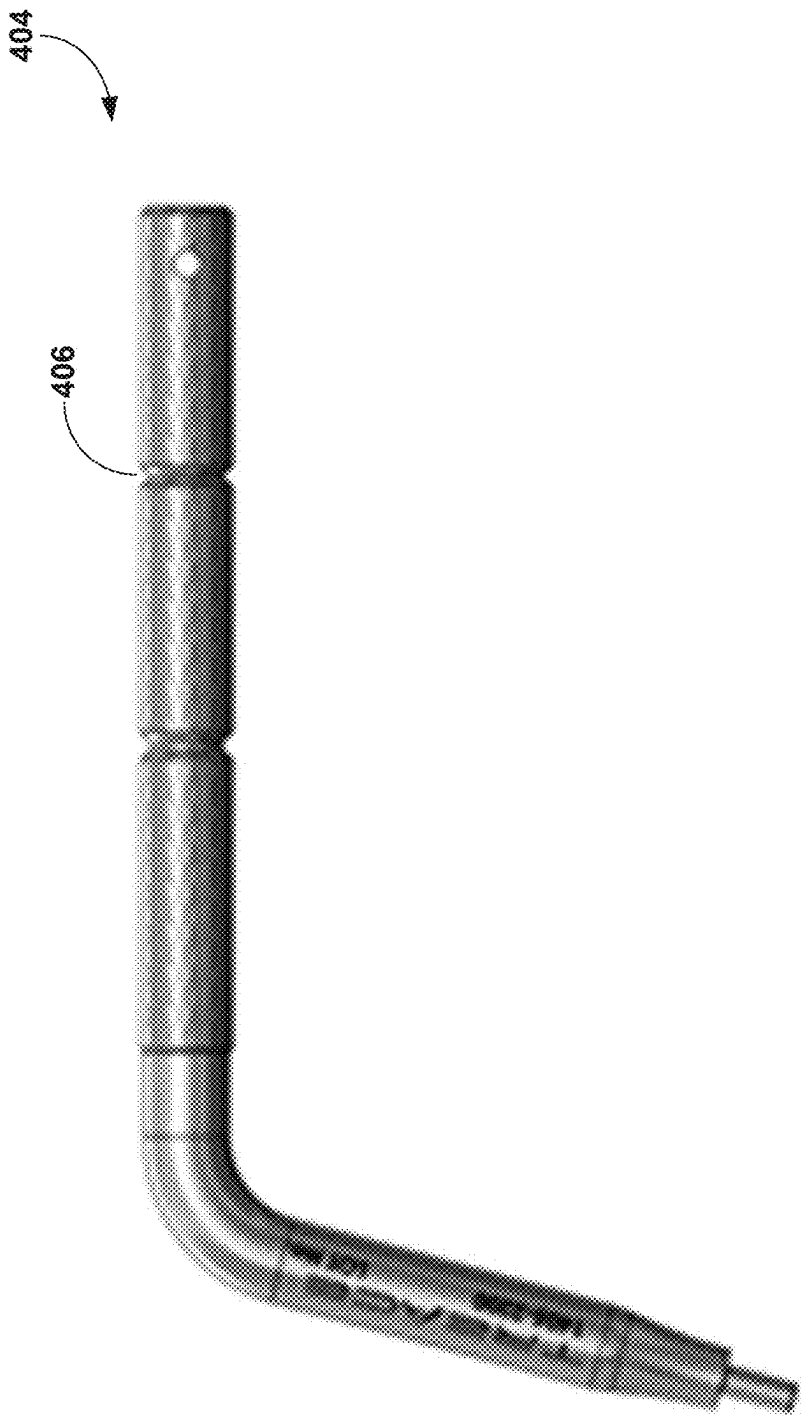
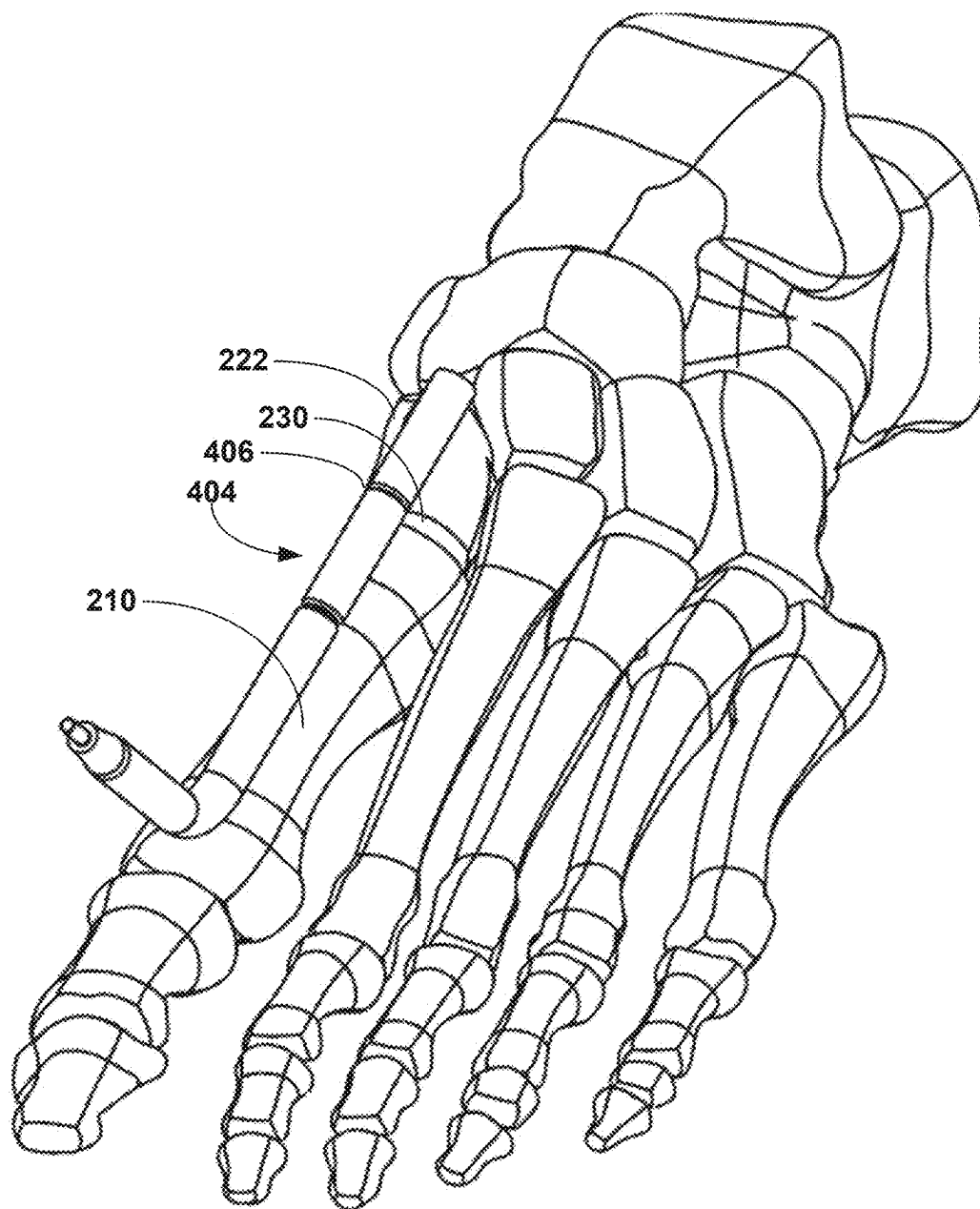


FIG. 6

**FIG. 7**

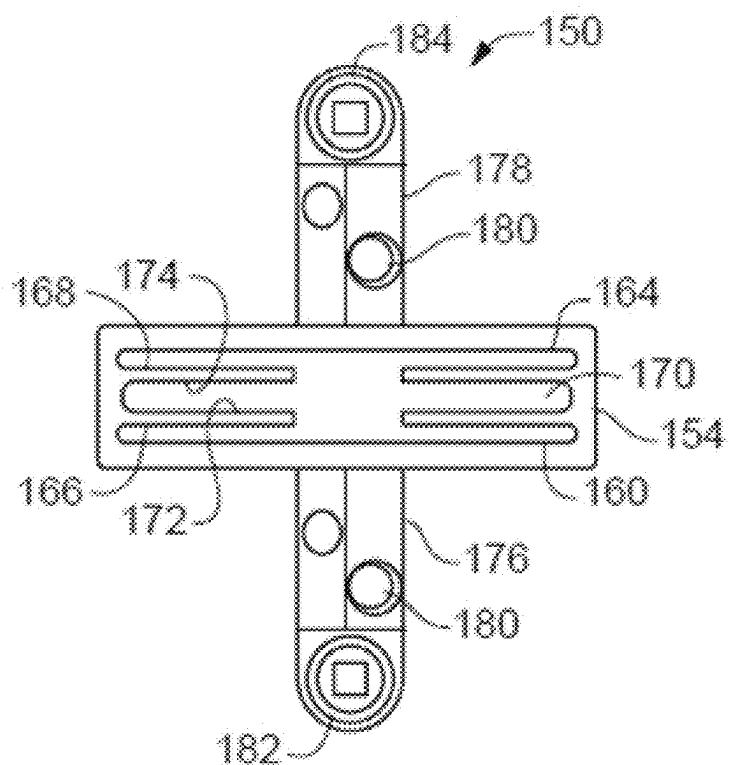
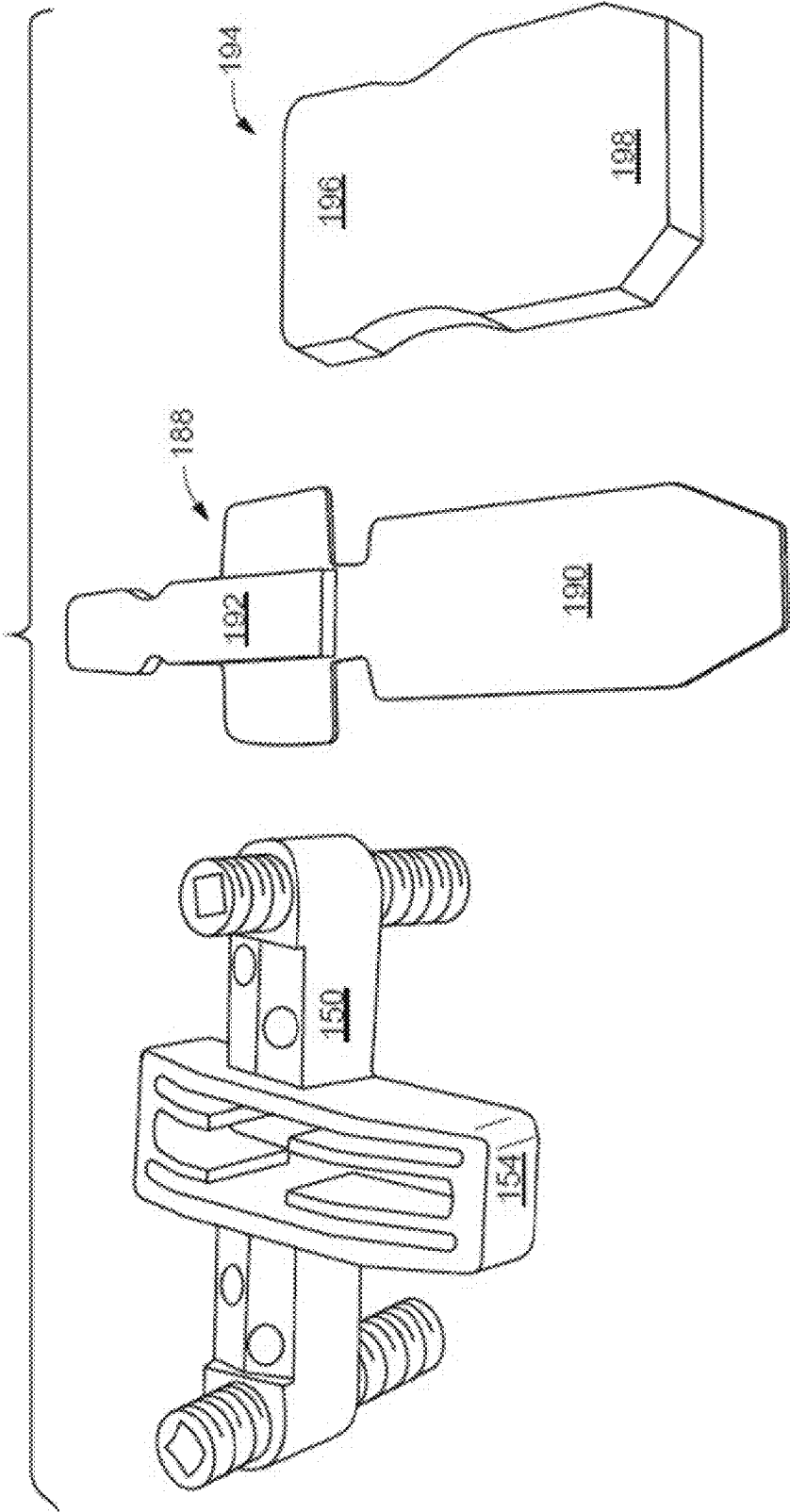


FIG. 8

FIG. 9



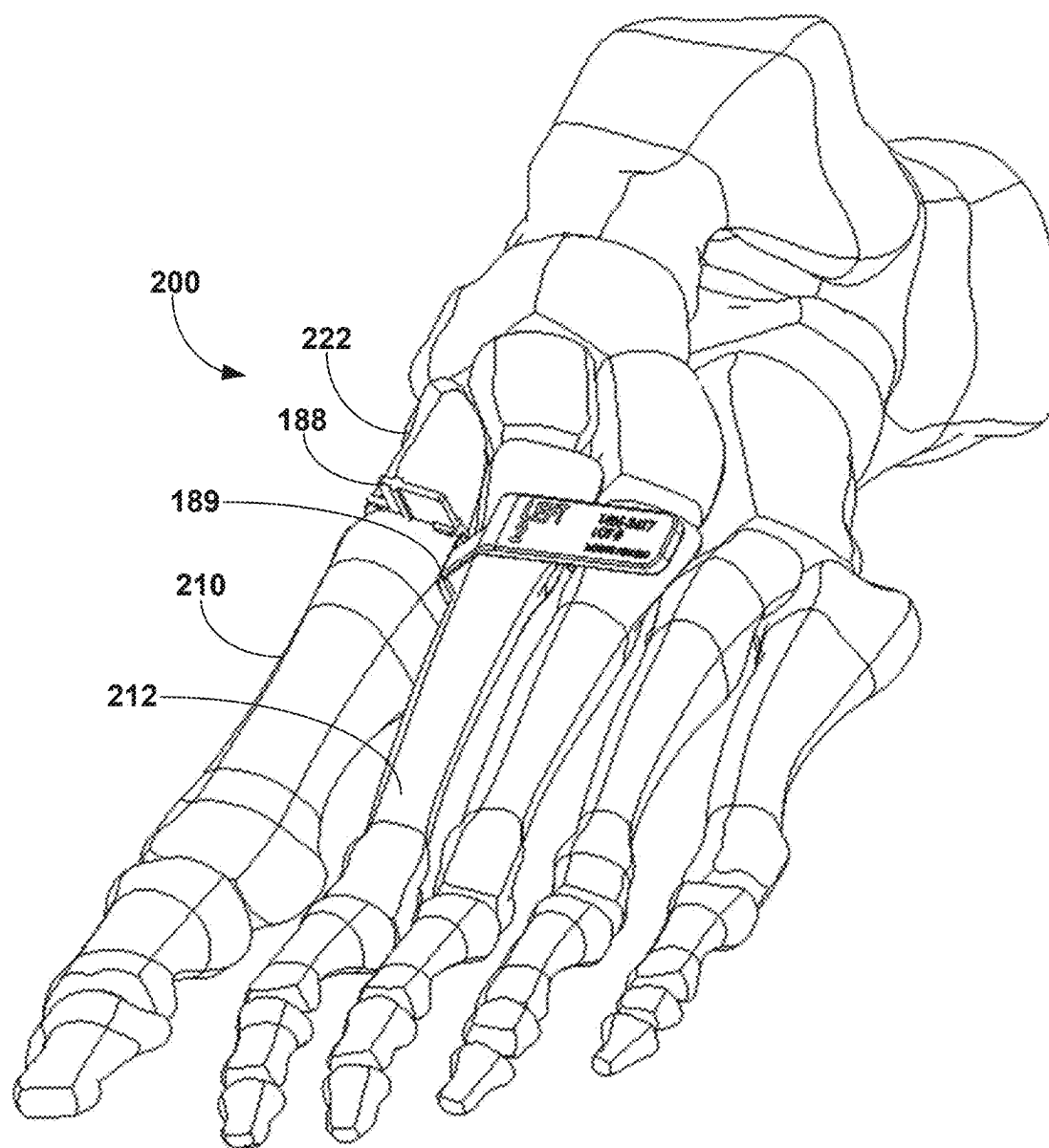


FIG. 10

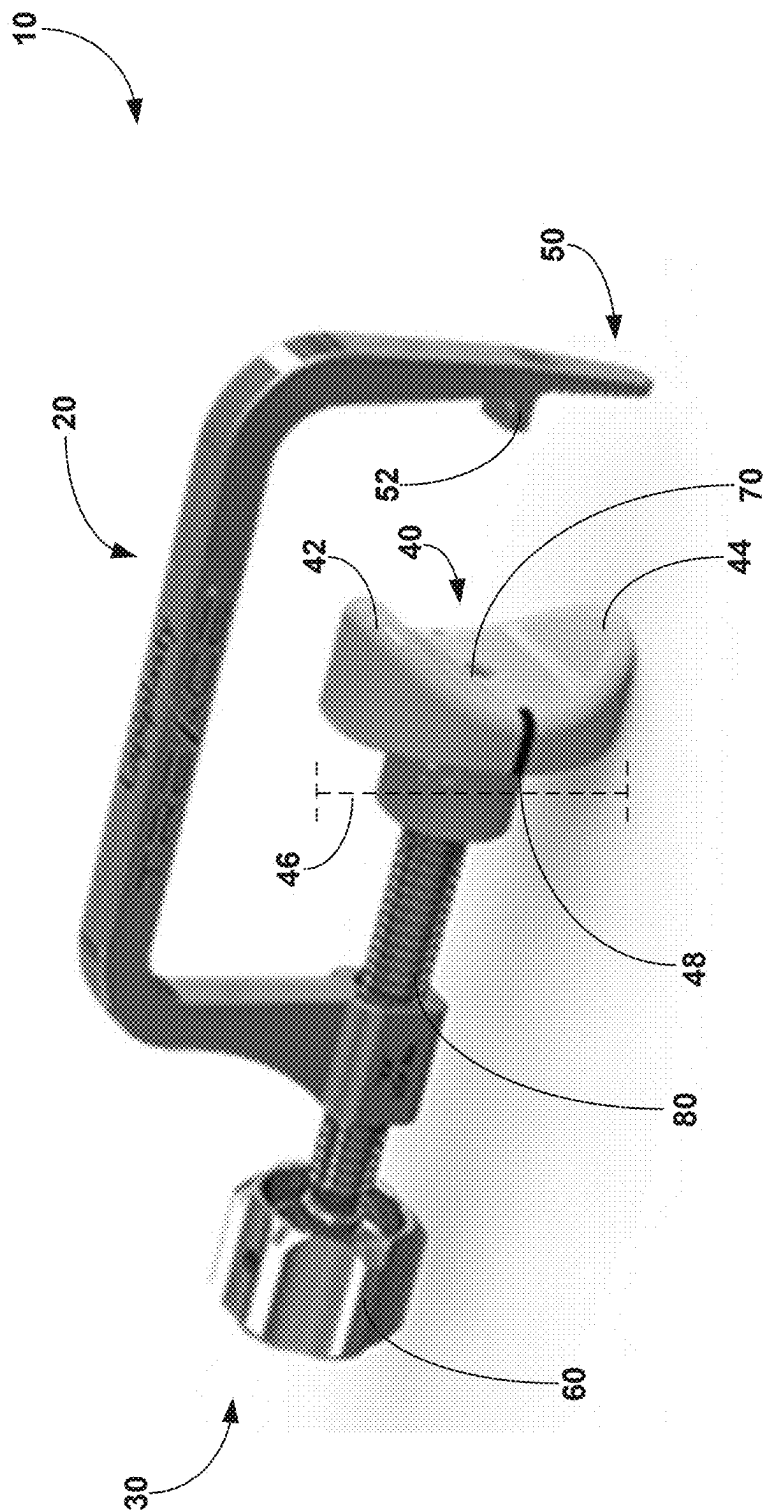


FIG. 11

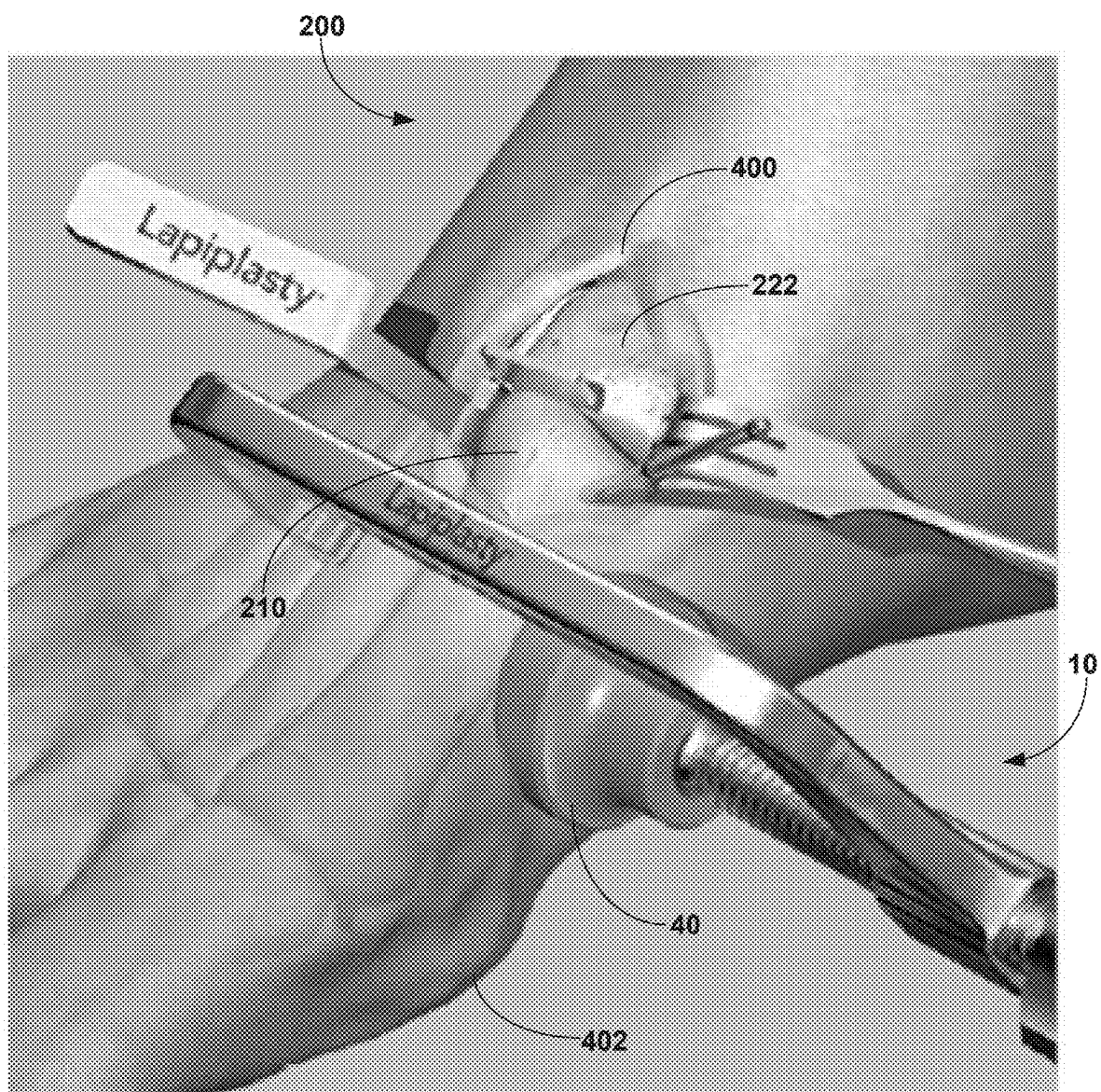


FIG. 12

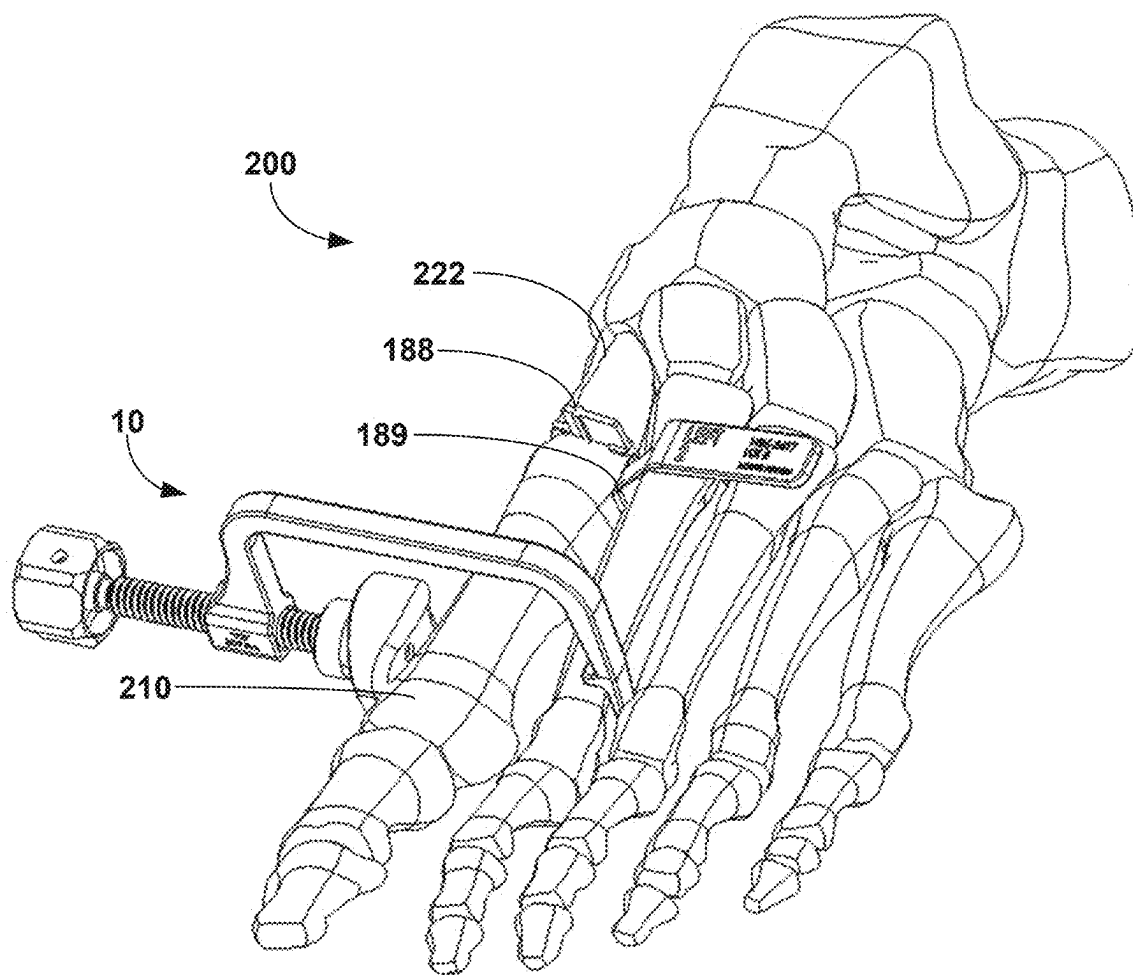


FIG. 13

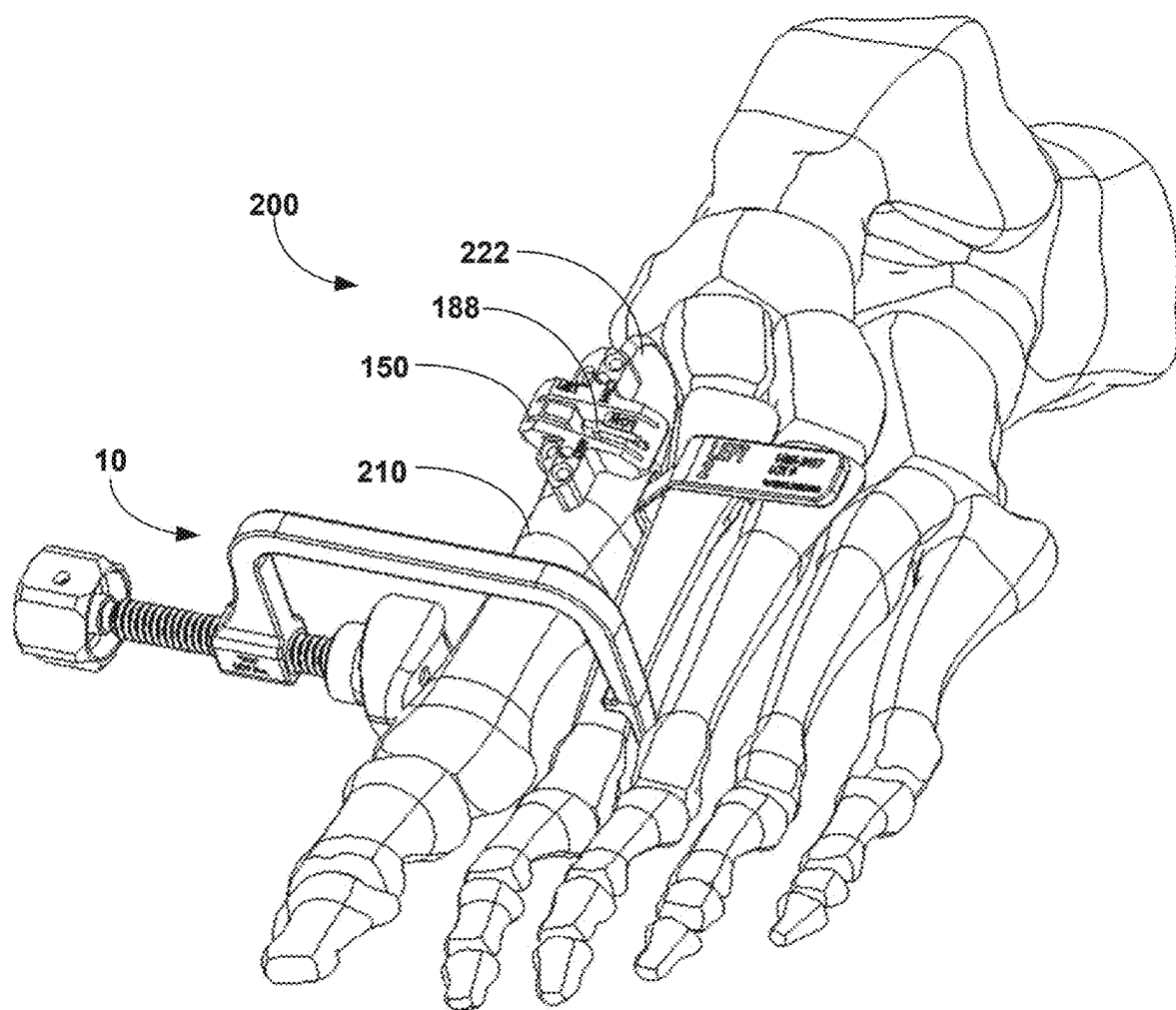


FIG. 14

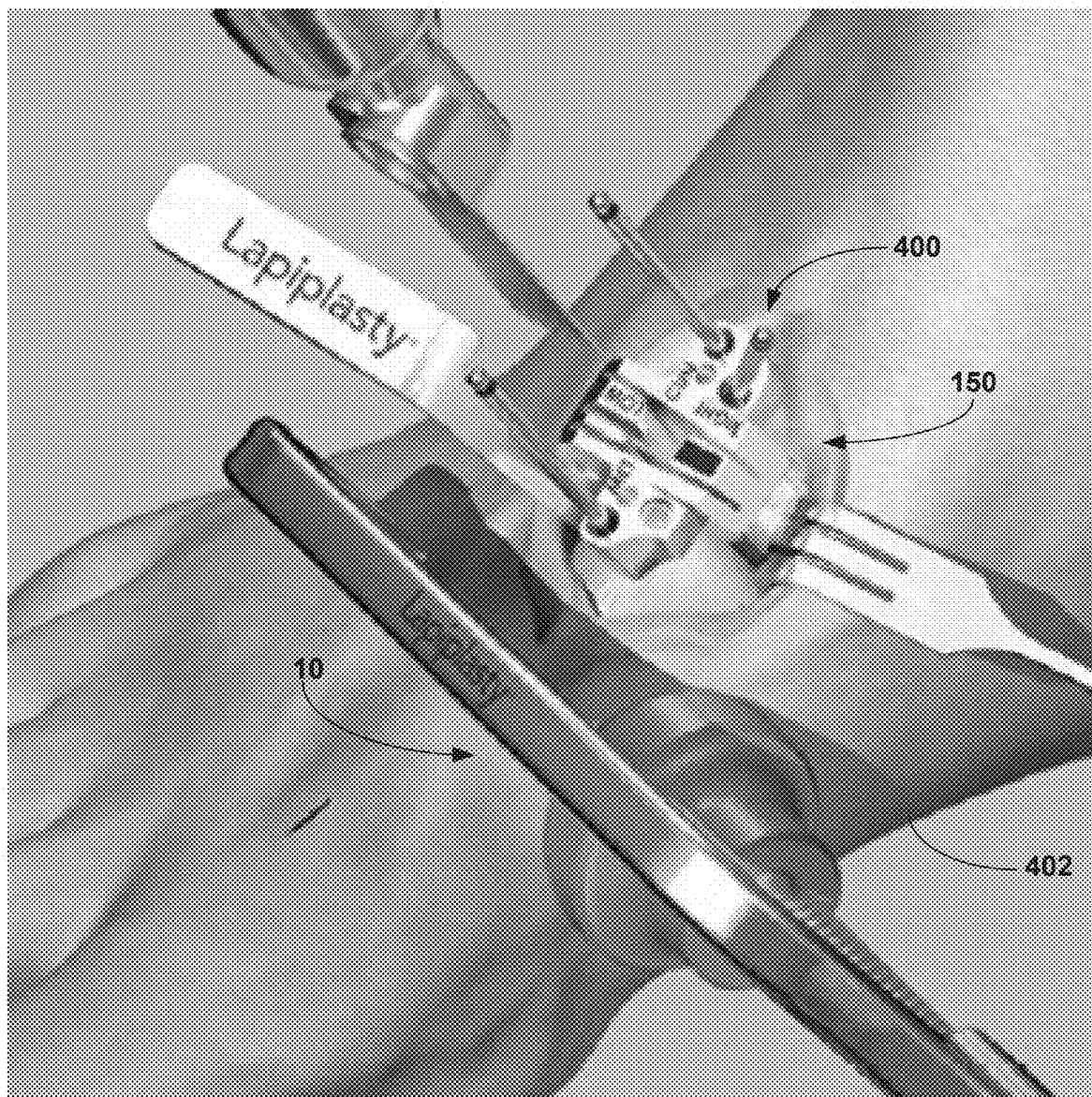


FIG. 15

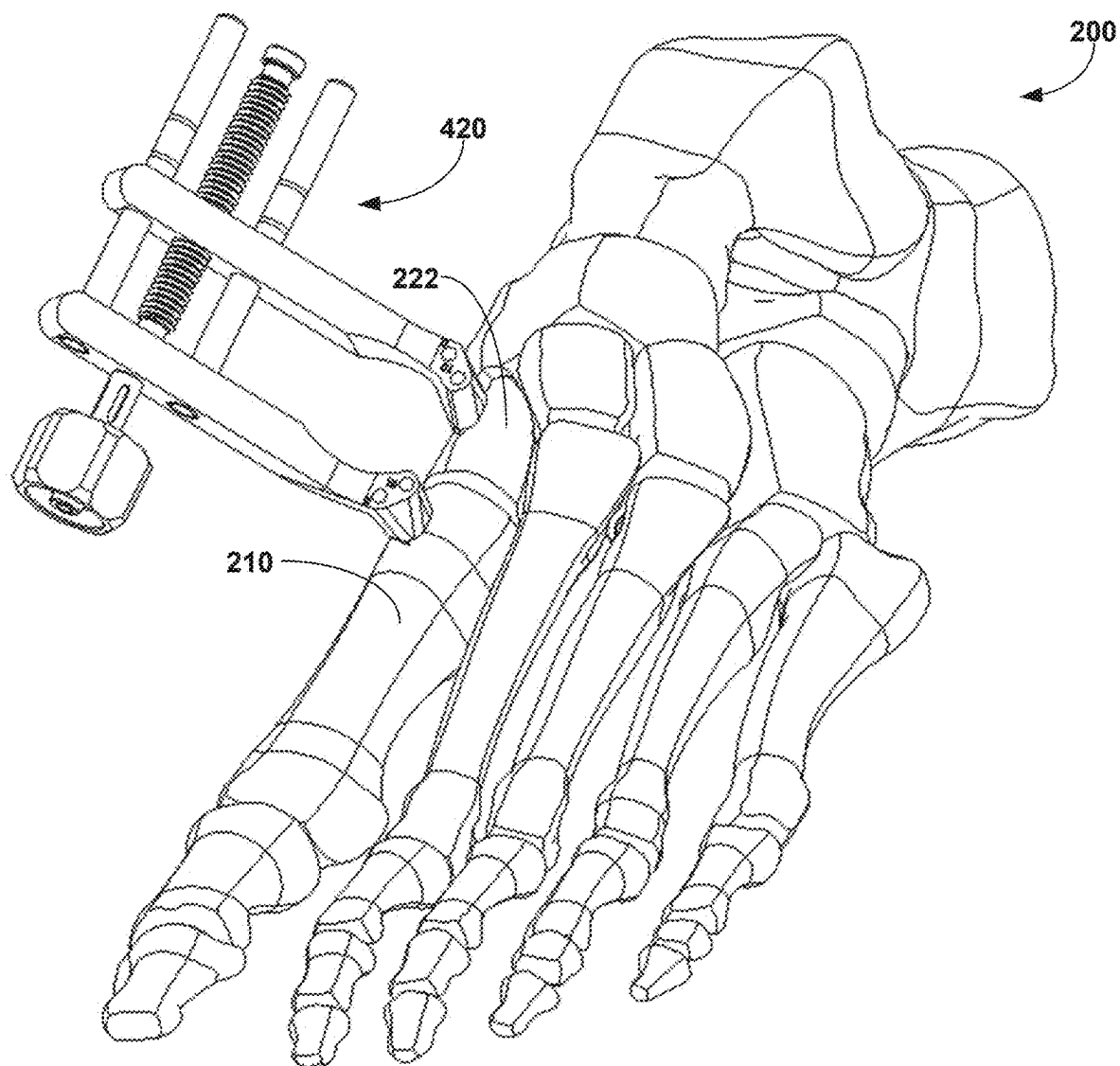


FIG. 16

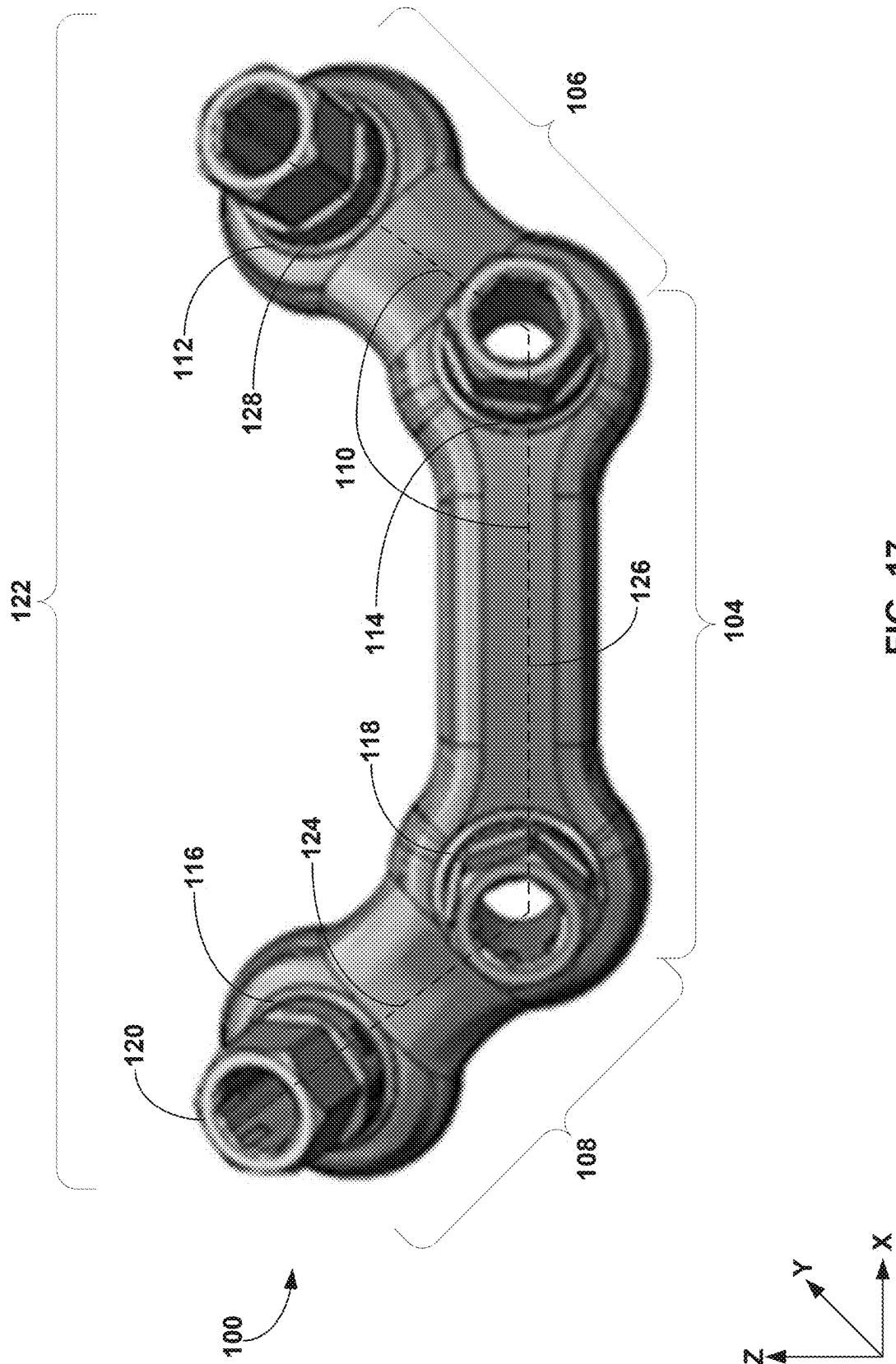


FIG. 17

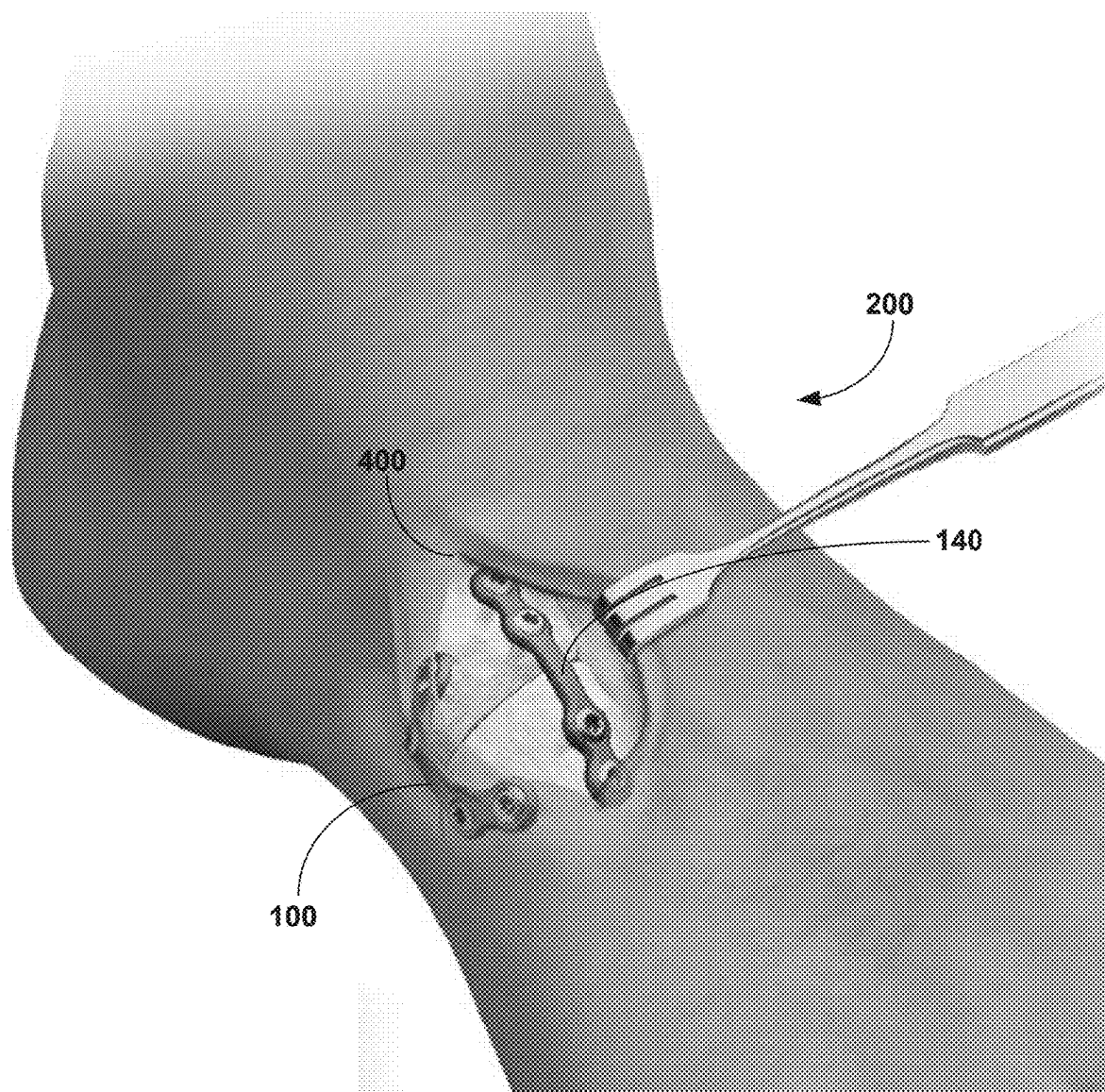


FIG. 18

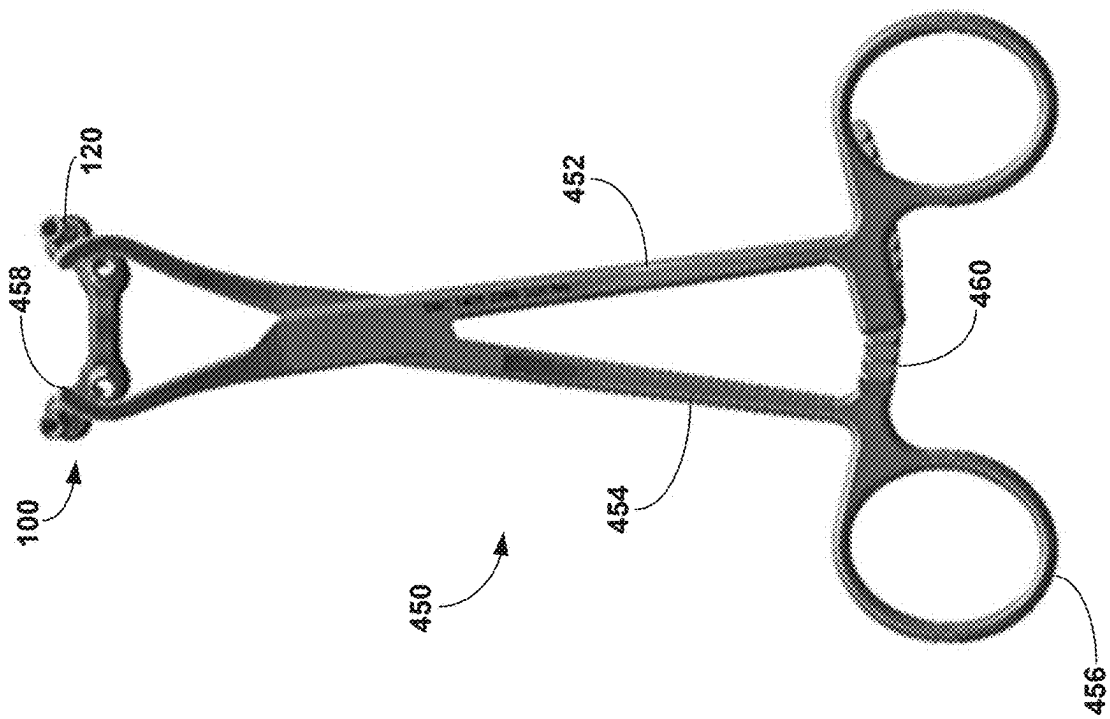


FIG. 19

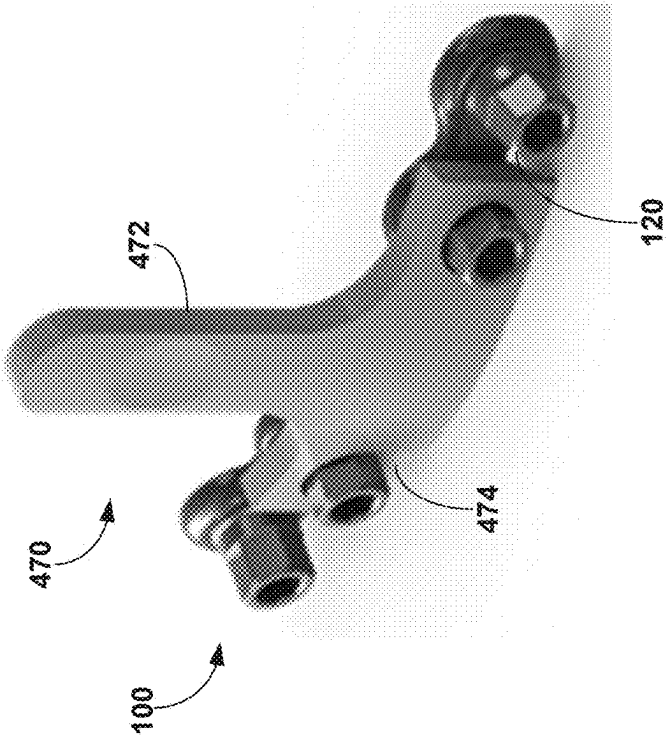


FIG. 20

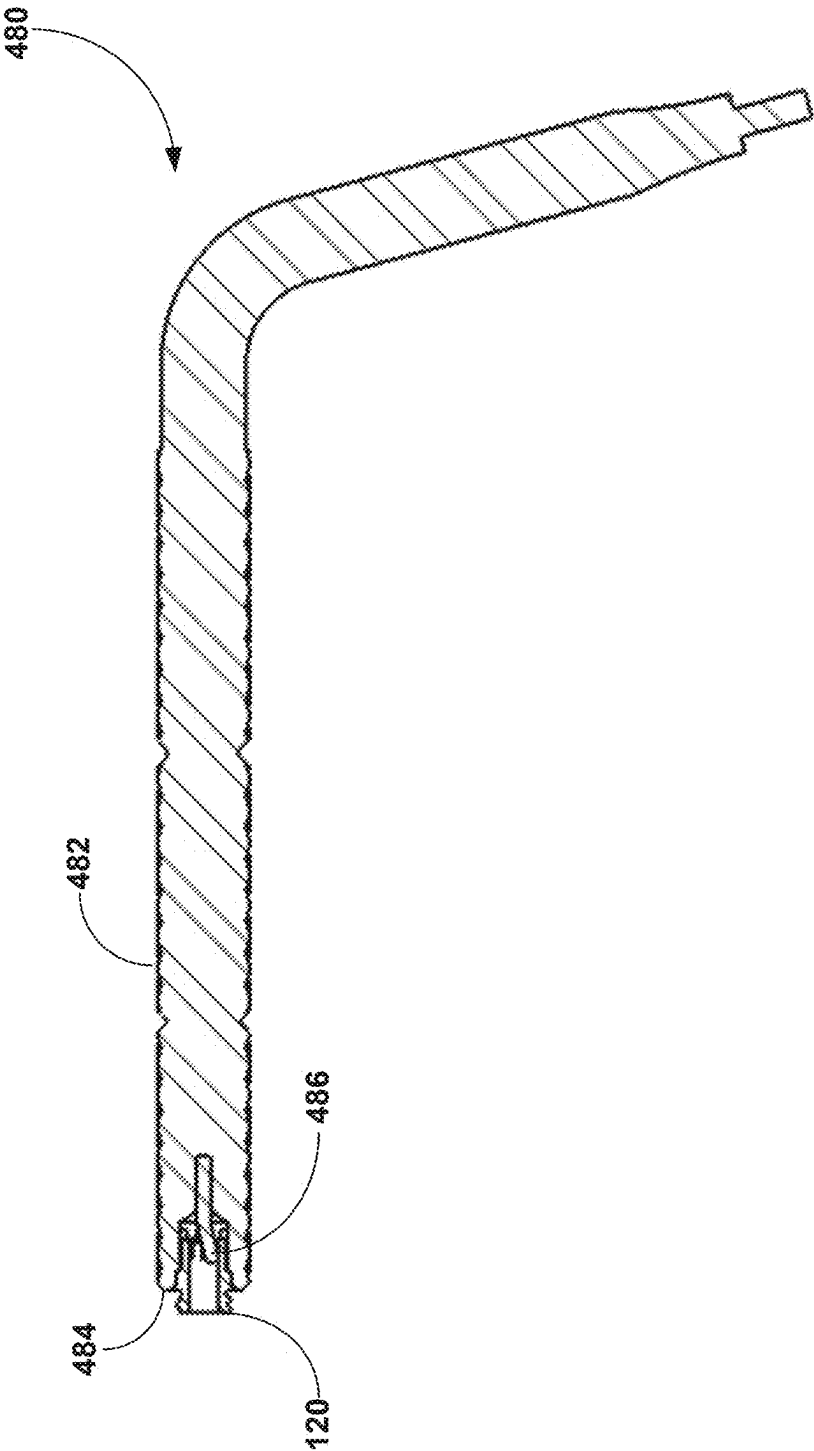


FIG. 21

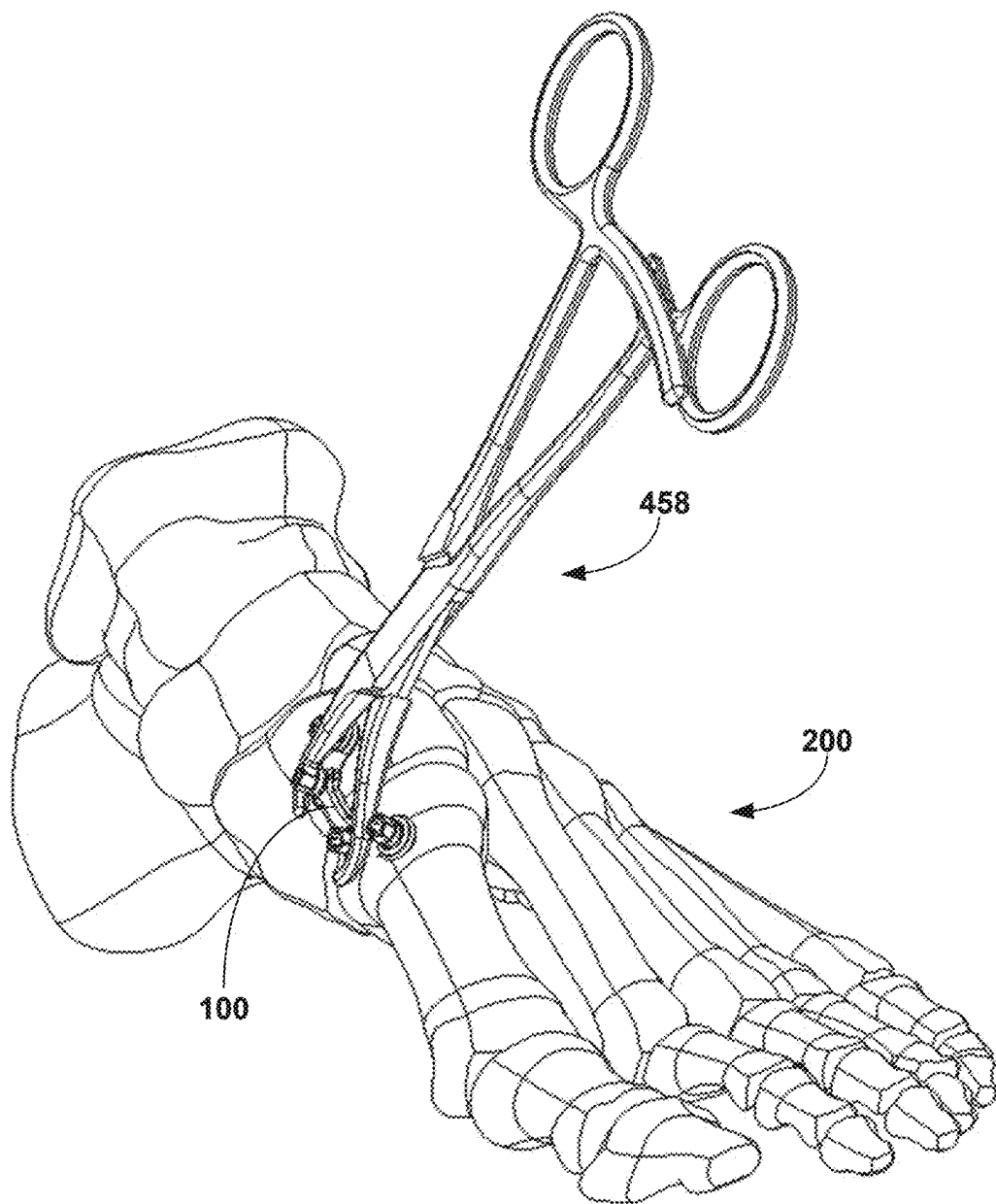


FIG. 22

SYSTEM AND TECHNIQUE FOR METATARSAL REALIGNMENT WITH REDUCED INCISION LENGTH

RELATED MATTERS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/674,447, filed Feb. 17, 2022, which claims the benefit of U.S. Provisional Patent Application No. 63/151,041, filed Feb. 18, 2021, the entire contents of each of these applications are hereby incorporated by reference.

TECHNICAL FIELD

[0002] This disclosure relates generally to devices and techniques for repositioning bones and, more particularly, to devices and techniques for repositioning bones in the foot.

BACKGROUND

[0003] Bones within the human body, such as bones in the foot, may be anatomically misaligned. For example, one common type of bone deformity is hallux valgus, which is a progressive foot deformity in which the first metatarsophalangeal joint is affected and is often accompanied by significant functional disability and foot pain. The metatarsophalangeal joint is laterally deviated, resulting in an abduction of the first metatarsal while the phalanges adduct. This often leads to development of soft tissue and a bony prominence on the medial side of the foot, which is called a bunion.

[0004] Surgical intervention may be used to correct a bunion deformity. A variety of different surgical procedures exist to correct bunion deformities and may involve removing the abnormal bony enlargement on the first metatarsal and/or attempting to realign the first metatarsal relative to the adjacent metatarsal. Surgical instruments that can facilitate efficient, accurate, and reproducible clinical results are useful for practitioners performing bone realignment techniques.

[0005] For patients, surgical intervention requires making one or more incisions through the patient's skin to access the underlying bones to perform a corrective procedure. A longer incision provides the surgeon with greater access to perform the procedure. However, a longer incision results in a longer scar for the patient after healing, which can be cosmetically undesirable. For this reason, the patient may prefer a shorter incision and/or fewer incisions. This can be challenging for the surgeon because it limits access for performing the corrective procedure.

SUMMARY

[0006] In general, this disclosure is directed to devices, systems, and techniques for performing a metatarsal realignment while minimizing the length and/or number of the incision(s) needed to access the bones for performing the procedure. As a result, a patient undergoing the procedure may have a shorter, less visible incision and resulting scar line, providing healing and/or cosmetic benefits as compared to a procedure performed using a longer incision and/or more incisions.

[0007] In some implementations, a clinician surgically accesses a tarsometatarsal ("TMT") joint defined between a metatarsal and an opposed cuneiform. The TMT joint may be the first TMT joint between the first metatarsal and medial cuneiform or a lesser TMT joint between a lesser

metatarsal and opposed cuneiform. In either case, the clinician can make a comparatively small incision to access the TMT joint. Once accessed, the clinician can prepare the end face of the metatarsal and the end face of the opposed cuneiform. Example preparation steps may include reaming, cutting, rongeur, curetting, burring, fenestrating and/or other similar techniques for exposing subchondral bone and/or establishing bleeding bone faces to promote fusion following rejoining of the metatarsal and the cuneiform.

[0008] Either before or after preparation of one or both end faces, the metatarsal is realigned within one or more planes in three-dimensional space. In one example, the clinician engages a bone positioner (also referred to as a bone positioning device) with the metatarsal and a bone other than the metatarsal. The bone positioner can then be actuated to move the metatarsal in one or more planes for realignment. The bone positioner may have a metatarsal engagement member positionable in contact with an external surface of the skin overlying the metatarsal. In other words, instead of making a separate incision to place the bone positioner in contact with the metatarsal bone, the bone positioner can engage the metatarsal bone through the skin to avoid the need for an additional incision.

[0009] In some examples, the bone positioner includes a tip that can be positioned on a lateral side of a metatarsal other than the metatarsal being moved. For example, where the metatarsal being moved is a first metatarsal, a small stab incision may be made on a lateral side of a metatarsal other than the metatarsal being moved, such as between the second metatarsal and the third metatarsal. In these examples, the tip of the bone positioner can be placed in contact with the lateral side of the second metatarsal. The bone positioning may include a mechanism that moves the metatarsal engagement member and the tip towards each other, causing the metatarsal to move in one or more planes to help correct a misalignment of the metatarsal. Additionally or alternatively, the clinician may grasp a pin inserted into the metatarsal to move the metatarsal in one or more planes.

[0010] After moving the metatarsal relative to the opposed cuneiform and/or an adjacent metatarsal, the clinician can perform a variety of steps to complete the surgical procedure. In some examples, the clinician installs a compressor and utilizes the compressor to compress the prepared end face of the metatarsal together with the prepared end face of the cuneiform. Additionally or alternatively, the clinician may temporarily fixate a moved position of the metatarsal relative to the cuneiform. For example, the clinician can insert one or more fixation wires through the metatarsal into one or more adjacent bones (e.g., an adjacent metatarsal, across the TMT joint into the opposed cuneiform).

[0011] In either case, clinician may cause the metatarsal to fuse to the cuneiform. For example, the clinician may install one or more fixation devices across the TMT joint to fixate the position of the metatarsal relative to the cuneiform for subsequent healing and fusion. In some implementations, the clinician installs one more bone plates across the TMT joint. For example, the clinician may utilize a generally U-shaped bone plate have a base and two outwardly extending legs. The clinician may position the base across the TMT joint with the legs extending upwardly in a dorsal direction relative to the base. This type of plate configuration can position one or more fixation holes of the bone plate at a location accessible by the clinician through the compara-

tively small incision. By contrast, when utilizing a straight plate or other plate configuration, one or more fixation holes of the bone plate may be located subdermally (after inserting the bone plate through the incision) in a way that can make it challenging for the clinician to install screws through the fixation holes.

[0012] In some implementations, the clinician may utilize a plate holder to help manipulate and/or hold the bone plate within the incision. The plate holder may be an instrument configured to releasably grasp an external surface of the plate and/or be inserted into a fixation hole of the plate, such as a drill guide inserted through the fixation hole. Using the plate holder, the clinician may be able to controllably and precisely position the bone plate and/or hold the bone plate during installation of one or more screws in the small incision created to access the TMT joint space. This can help address space constraints caused by the comparatively small length incision, which can make it difficult for the clinician to properly position the bone plate using their fingers alone.

[0013] In one example, a method of performing a minimal incision metatarsal correction procedure is described. The method includes preparing an end of a metatarsal and preparing an end of a cuneiform separated from the metatarsal by a tarsometatarsal joint. The method also involves moving the metatarsal relative to the cuneiform using a bone positioning device that includes a metatarsal engagement member and a tip. The example specifies that the metatarsal engagement member is positioned in contact with a skin of the patient covering a medial side of the metatarsal and that the tip is positioned on a lateral side of a different metatarsal. The method also includes, after moving the metatarsal relative to the cuneiform, causing the metatarsal to fuse to the cuneiform.

[0014] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF DRAWINGS

[0015] FIGS. 1A and 1B are front views of a foot showing a normal first metatarsal position and an example frontal plane rotational misalignment position, respectively.

[0016] FIGS. 2A and 2B are top views of a foot showing a normal first metatarsal position and an example transverse plane misalignment position, respectively.

[0017] FIGS. 3A and 3B are side views of a foot showing a normal first metatarsal position and an example sagittal plane misalignment position, respectively.

[0018] FIG. 4 is a flow diagram illustrating an example technique for performing a minimally invasive surgical procedure to correct a position of a metatarsal relative to an adjacent bone.

[0019] FIG. 5 is a perspective view of a foot showing example surgical access to a TMT joint.

[0020] FIG. 6 is an image of one example configuration of an incision guide that can be used to help designate the location of a TMT joint prior to making an incision.

[0021] FIG. 7 is a perspective view of a foot showing the incision guide of FIG. 6 positioned over a metatarsal with a radio-identifiable marking line.

[0022] FIG. 8 illustrates one example bone preparation guide that may be used as part of a minimally invasive procedure.

[0023] FIG. 9 illustrates example instrumentation that may be used with the example bone preparation guide of FIG. 8.

[0024] FIG. 10 is a perspective view of an example foot in which a bi-planar instrument is positioned in adjacent joint spaces.

[0025] FIG. 11 shows a side perspective view of an example bone positioner that can be used to move a metatarsal relative to an adjacent bone.

[0026] FIG. 12 is an illustration showing an example arrangement of the bone positioning device of FIG. 11 in which a metatarsal engagement member is positioned in contact with the external surface of the skin of the patient covering a side of a metatarsal.

[0027] FIG. 13 is a perspective view of a foot showing a seeker and a fulcrum inserted between adjacent joint spaces and a bone positioner engaged with a first metatarsal.

[0028] FIG. 14 is a perspective view of the foot from FIG. 13 further illustrated with a bone preparation guide installed on the seeker.

[0029] FIG. 15 is a further illustration of the surgical instrument arrangement of FIG. 14 showing the example location of an incision and skin of the patient relative to the surgical instruments.

[0030] FIG. 16 is a perspective view of a foot illustrating an example compressing instrument that may be used to compress the prepared end faces of the metatarsal and medial cuneiform together.

[0031] FIG. 17 is illustration of an example bone plate that can be used to help permanently fixate a moved position of a metatarsal relative to an opposed cuneiform.

[0032] FIG. 18 is a perspective view of a foot showing an example insertion location of the bone plate of FIG. 17.

[0033] FIGS. 19-21 are illustrations of example plate holder configurations that may be used to manipulate a bone plate during a minimal incision procedure.

[0034] FIG. 22 is an illustration of a foot showing an example plate holder manipulating a plate for placement across a TMT joint.

DETAILED DESCRIPTION

[0035] This disclosure generally relates to devices, systems, and techniques for performing a minimal incision bone realignment procedure. In an exemplary application, the devices and techniques can be used during a surgical procedure performed on one or more bones, such as a bone alignment, osteotomy, fusion procedure, fracture repair, and/or other procedures where one or more bones are to be set in a desired position. Such a procedure can be performed, for example, on bones (e.g., adjacent bones separated by a joint or different portions of a single bone) in the foot or hand, where bones are relatively small compared to bones in other parts of the human anatomy. In one example, a procedure utilizing an embodiment of the disclosure can be performed to correct an alignment between a metatarsal (e.g. a first metatarsal) and a cuneiform (e.g., a medial cuneiform), such as a bunion correction. An example of such a procedure is a lapidus procedure. In another example, the procedure can be performed by modifying an alignment of a metatarsal (e.g. a first metatarsal). An example of such a procedure is a basilar metatarsal osteotomy procedure.

[0036] Preparation and fusion of two opposed bone portions, such as a metatarsal and cuneiform, may be performed according to the disclosure for a variety of clinical reasons and indications. Preparation and fusion of a metatarsal and

cuneiform at the TMT joint may be performed to treat hallux valgus and/or other bone and/or joint conditions.

[0037] Hallux valgus, also referred to as hallux abducto valgus, is a complex progressive condition that is characterized by lateral deviation (valgus, abduction) of the hallux and medial deviation of the first metatarsophalangeal joint. Hallux valgus typically results in a progressive increase in the hallux abductus angle, the angle between the long axes of the first metatarsal and proximal phalanx in the transverse plane. An increase in the hallux abductus angle may tend to laterally displace the plantar aponeurosis and tendons of the intrinsic and extrinsic muscles that cross over the first metatarsophalangeal joint from the metatarsal to the hallux. Consequently, the sesamoid bones may also be displaced, e.g., laterally relative to the first metatarsophalangeal joint, resulting in subluxation of the joints between the sesamoid bones and the head of the first metatarsal. This can increase the pressure between the medial sesamoid and the crista of the first metatarsal head.

[0038] While techniques and devices are described herein particularly in connection with the first metatarsal and medial cuneiform of the foot, the techniques and devices may be used on other adjacent bones separated by a joint in the hand or foot. For example, the techniques and devices may be performed on a different metatarsal (e.g., second, third, fourth, or fifth metatarsal) and its opposed cuneiform/cuboid. Accordingly, reference to a first metatarsal and medial cuneiform herein may be replaced with other bone pairs without departing from the scope of the disclosure.

[0039] To further understand example techniques of the disclosure, the anatomy of the foot will first be described with respect to FIGS. 1-3 along with example misalignments that may occur and be corrected according to the present disclosure. A bone misalignment may be caused by hallux valgus (bunion), a natural growth deformity, and/or other condition.

[0040] FIGS. 1A and 1B are front views of foot 200 showing a normal first metatarsal position and an example frontal plane rotational misalignment position, respectively. FIGS. 2A and 2B are top views of foot 200 showing a normal first metatarsal position and an example transverse plane misalignment position, respectively. FIGS. 3A and 3B are side views of foot 200 showing a normal first metatarsal position and an example sagittal plane misalignment position, respectively. While FIGS. 1B, 2B, and 3B show each respective planar misalignment in isolation, in practice, a metatarsal may be misaligned in any two of the three planes or even all three planes. Accordingly, it should be appreciated that the depiction of a single plane misalignment in each of FIGS. 1B, 2B, and 3B is for purposes of illustration and a metatarsal may be misaligned in multiple planes that is desirably corrected. Further, a bone condition treated according to the disclosure may not present any of the example misalignments described with respect to FIGS. 1B, 2B, and 3B, and it should be appreciated that the disclosure is not limited in this respect.

[0041] With reference to FIGS. 1A and 2A, foot 200 is composed of multiple bones including a first metatarsal 210, a second metatarsal 212, a third metatarsal 214, a fourth metatarsal 216, and a fifth metatarsal 218. The metatarsals are connected distally to phalanges 220 and, more particularly, each to a respective proximal phalanx. The first metatarsal 210 is connected proximally to a medial cuneiform 222, while the second metatarsal 212 is connected

proximally to an intermediate cuneiform 224 and the third metatarsal is connected proximally to lateral cuneiform 226. The fourth and fifth metatarsals 216, 218 are connected proximally to the cuboid bone 228. The joint 230 between a metatarsal and respective cuneiform (e.g., first metatarsal 210 and medial cuneiform 222) is referred to as the tarso-metatarsal ("TMT") joint. The joint 232 between a metatarsal and respective proximal phalanx is referred to as a metatarsophalangeal joint. The angle 234 between adjacent metatarsals (e.g., first metatarsal 210 and second metatarsal 212) is referred to as the intermetatarsal angle ("IMA").

[0042] As noted, FIG. 1A is a frontal plane view of foot 200 showing a typical position for first metatarsal 210. The frontal plane, which is also known as the coronal plane, is generally considered any vertical plane that divides the body into anterior and posterior sections. On foot 200, the frontal plane is a plane that extends vertically and is perpendicular to an axis extending proximally to distally along the length of the foot. FIG. 1A shows first metatarsal 210 in a typical rotational position in the frontal plane. FIG. 1B shows first metatarsal 210 with a frontal plane rotational deformity characterized by a rotational angle 236 relative to ground, as indicated by line 238.

[0043] FIG. 2A is a top view of foot 200 showing a typical position of first metatarsal 210 in the transverse plane. The transverse plane, which is also known as the horizontal plane, axial plane, or transaxial plane, is considered any plane that divides the body into superior and inferior parts. On foot 200, the transverse plane is a plane that extends horizontally and is perpendicular to an axis extending dorsally to plantarly (top to bottom) across the foot. FIG. 2A shows first metatarsal 210 with a typical IMA 234 in the transverse plane. FIG. 2B shows first metatarsal 210 with a transverse plane rotational deformity characterized by a greater IMA caused by the distal end of first metatarsal 210 being pivoted medially relative to the second metatarsal 212.

[0044] FIG. 3A is a side view of foot 200 showing a typical position of first metatarsal 210 in the sagittal plane. The sagittal plane is a plane parallel to the sagittal suture which divides the body into right and left halves. On foot 200, the sagittal plane is a plane that extends vertically and is perpendicular to an axis extending proximally to distally along the length of the foot. FIG. 3A shows first metatarsal 210 with a typical rotational position in the sagittal plane. FIG. 3B shows first metatarsal 210 with a sagittal plane rotational deformity characterized by a rotational angle 240 relative to ground, as indicated by line 238.

[0045] Surgical techniques and instruments according to the disclosure can be useful to correct a misalignment of one or more bones, such as the metatarsal and opposed cuneiform, and/or promote fusion of the metatarsal and cuneiform across the TMT joint. In some applications, the technique involves forming two incisions: a comparatively small incision to access the TMT joint and a second stab incision between two metatarsals (e.g., between a second metatarsal and a third metatarsal). The end of the metatarsal and the end of the cuneiform can be prepared. For example, the clinician can insert a bone preparation instrument through the comparatively small incision to prepare the end face of one or both bones.

[0046] Before or after preparing one or both ends, a bone positioner having a metatarsal engagement member and tip can be installed to help facilitate realignment of the metatarsal relative to the cuneiform. The metatarsal engagement

member can be positioned in contact with the external surface of the skin of the patient overlaying the medial side of the metatarsal. The tip can be inserted through the small stab incision between the adjacent metatarsals. For example, the tip can be inserted through the small stab incision and the tip positioned in contact with a lateral face of a metatarsal different than the metatarsal being moved. In either case, the bone positioner may include one more mechanisms that allow the metatarsal engagement member to be moved relative to the tip, resulting in the metatarsal moving in one or more planes relative to the cuneiform.

[0047] With the metatarsal suitably repositioned, the clinician may temporarily and/or permanently fixate the moved position of the metatarsal relative to the cuneiform. For example, the clinician may insert a fixation pin through the metatarsal and into the cuneiform (e.g., across the TMT joint) to hold the position of the metatarsal for subsequent installation of one or more permanent fixation devices. In some implementations, the clinician compresses the prepared end face of the metatarsal together with the prepared end face of the cuneiform before installing one or more temporary and/or permanent fixation devices.

[0048] In either case, in some examples, the clinician may grasp a plate holder instrument operatively connected to a bone plate and use the instrument to position the bone plate through the comparatively small incision. Through manipulation of the plate holder instrument, the clinician may position and hold the bone plate across the TMT joint while the clinician inserts one or more fixation elements through one or more corresponding fixation holes in the bone plate into the underlying metatarsal and/or cuneiform.

[0049] FIG. 4 is a flow diagram illustrating an example technique for performing a minimally invasive metatarsal realignment procedure. The example technique will be described with respect to first metatarsal 210 and medial cuneiform 222, although can be performed on other bones, as discussed above. For purposes of discussion, the technique of FIG. 4 will be discussed with respect to different example images illustrated in FIGS. 5-22.

[0050] FIG. 5 is a perspective view of a foot 200 showing example surgical access to the TMT joint. With reference to FIGS. 4 and 5, the example technique involves surgically accessing TMT joint 230 separating first metatarsal 210 from opposed medial cuneiform 222 (300). To surgically access the joint, the patient may be placed in a supine position on the operating room table and general anesthesia or monitored anesthesia care administered. Hemostasis can be obtained by applying thigh tourniquet or mid-calf tourniquet. An incision 400 (FIG. 5) can be made through the skin 402, such as on a dorsal side of the foot, a medial side of the foot, or on a dorsal-medial side of the foot.

[0051] To help identify the proper location to make incision 400, in some implementations, the clinician may image foot 200 prior to making the incision. For example, the clinician may take a fluoroscopic (e.g. X-ray) image of at least a portion of foot 200 encompassing TMT joint 230 prior to making incision 400. Imaging foot 200 can reveal to the clinician the location of TMT joint 230 about which incision 400 can be centered when subsequently cutting through skin 402. In these applications, the clinician may use a tool such as a ruler, incision guide, or other instrument fabricated at least partially from a radiopaque material to designate the location of TMT joint 230 under imaging.

[0052] FIG. 6 is an image of one example configuration of an incision guide 404 that can be used to help designate the location of TMT joint 230 prior to making incision 400. In this example, incision guide 404 is shown as including one or more radio-identifiable marking lines 406, which are distinguishable from a remainder of the incision guide under imaging. The one or more radio-identifiable marking lines 406 can be formed from a different material than a remainder of the incision guide, have a different thickness than a remainder of the incision guide, and/or otherwise be distinguishable under imaging from the remainder of the incision guide. In either configuration, the clinician may align radio-identifiable marking line 406 with TMT joint 230 under imaging to designate the location for subsequently making incision 400.

[0053] FIG. 7 is a perspective view of foot 200 showing incision guide 404 positioned over metatarsal 210 with radio-identifiable marking line 406 generally co-planar with TMT joint 230 between the metatarsal and medial cuneiform 222. By accurately identifying the location of TMT joint 230 prior to making incision 400, the clinician may make a comparatively smaller incision substantially centered on the TMT joint than if the clinician makes an exploratory incision without first identifying the precise location of the joint.

[0054] Independent of whether the clinician does or does not utilize an imaging device to identify TMT joint 230 prior to making incision 400, the clinician can make the incision to access the underlying joint. While the specific length of incision 400 (in the distal to proximal direction parallel to the long axis of first metatarsal 210) may vary depending on the anatomy of the specific patient undergoing the procedure and the clinician performing the surgery, in some examples, the incision may have a length less than 12 cm. For example, the clinician may cut incision 400 to have a length less than 10.5 cm, less than 8 cm, less than 6 cm, or less than 5 cm. In some instances, incision 400 may have a length ranging from 1 cm to 12 cm, such as from 1.5 cm to 8 cm, 2 cm to 6 centimeters, or from 3 cm to 4 cm.

[0055] With TMT joint 230 exposed, the technique of FIG. 4 involves preparing the end face of first metatarsal 210 and/or preparing the end face of medial cuneiform 222 (302, 304). While FIG. 4 schematically illustrates an example order in which the end face of the first metatarsal and the end face of the medial cuneiform can be prepared, it should be appreciated that the surgical technique is not limited to any particular order of preparation. For example, the clinician can prepare the end face of either the metatarsal or the cuneiform before preparing the end face of the other bone. Further, one or both of the end faces of the metatarsal and the cuneiform can be prepared before and/or after the metatarsal is moved relative to the cuneiform. Accordingly, unless otherwise specified, the order of bone preparation and/or movement is not limited.

[0056] In general, the clinician can prepare the end of each bone forming a TMT joint so as to promote fusion of the bone ends across the TMT joint following realignment. Bone preparation may involve using a tissue removing instrument to apply a force to the end face of the bone so as to create a bleeding bone face to promote subsequent fusion. Example tissue removing instruments that can be used include, but are not limited to, a saw, a rotary bur, a rongeur, a reamer, an osteotome, a curette, and the like. The tissue removing instrument can be applied to the end face of the bone being prepared to remove cartilage and/or bone. For

example, the tissue removing instrument may be applied to the end face to remove cartilage (e.g., all cartilage) down to subchondral bone. Additionally or alternatively, the tissue removing instrument may be applied to cut, fenestrate, morselize, and/or otherwise reshape the end face of the bone and/or form a bleeding bone face to promote fusion. In instances where a cutting operation is performed to remove an end portion of a bone, the cutting may be performed freehand or with the aid of a cutting guide having a guide surface positionable over the portion of bone to be cut. When using a bone preparation guide, a cutting instrument can be inserted against the guide surface (e.g., between a slot define between two guide surfaces) to guide the cutting instrument for bone removal.

[0057] In some examples, the clinician cuts at least one bone defining the TMT joint **230** (e.g., one or both of first metatarsal **210** and medial cuneiform **222**). The clinician may cut both bones defining the TMT joint or may cut only one bone defining the joint and perform a different preparation technique on the other bone.

[0058] While a variety of different bone preparation guides can be used to guide a cutting instrument, FIG. **8** illustrates one example bone preparation guide **150** that may be used as part of a minimally invasive procedures. In the illustrated example, bone preparation guide **150** includes a body **154** defining at least one guide surface for guiding a cutting instrument which, in the illustrated example, is shown as including two guide surfaces: a first guide surface **160** to define a first preparing plane and a second guide surface **164** to define a second preparing plane. A tissue removing instrument (e.g., a saw, rotary bur, osteotome, etc.) can be aligned with and/or guided along (e.g., in contact with) the surface(s) to remove tissue (e.g., remove cartilage or bone and/or make cuts to bone). The first and second guide surfaces **160**, **164** can be spaced from each other by a distance, (e.g., between about 2 millimeters and about 10 millimeters, such as between about 4 and about 7 millimeters). In the embodiment shown, the first and second guide surfaces are parallel, such that cuts to adjacent bones using the guide surfaces will be generally parallel. However, in other examples, one guide surface can be angled relative to another guide surface at a non-zero degree angle.

[0059] In some configurations, as shown in FIG. **8**, a first facing surface **166** is positioned adjacent the first guide surface **160** and/or a second facing surface **168** is positioned adjacent the second guide surface **164**. In such configurations, the distance between the first guide surface and the first facing surface defines a first guide slot, and the distance between the second guide surface and the second facing surface defines a second guide slot. Each slot can be sized to receive a tissue removing instrument to prepare the bone ends. The first and second slots may be parallel or skewed. In the illustrated example, the facing surfaces each contain a gap, such that the surface is not a single, continuous surface. In other embodiments, one or both facing surfaces can be a single, continuous surface lacking any such gap.

[0060] The space between adjacent guide surfaces and/or cutting slots (when configured with multiple of such features) can be solid material or can include one or more openings. In the illustrated example, an opening **170** is defined by body **154** between the first and second guide surfaces. The opening can be an area between the guide surfaces useful for allowing a practitioner to have a visual path to bones during bone preparation and/or to receive an

instrument. In the configuration shown, the opening extends across the body and a distance from a surface **172** opposite of the first facing surface **166** to a surface **174** opposite of the second facing surface **168**.

[0061] The illustrated bone preparation guide also includes a first end **176** extending from the body **154** in a first direction and a second end **178** extending from the body in a second direction. The second direction can be different than the first direction (e.g., an opposite direction). As shown, each of the first end and the second end can include at least one fixation aperture **180** configured to receive a fixation pin to secure the bone preparation guide to an underlying bone. For example, first end **176** of bone preparation guide **150** may define a first fixation aperture through which a first pin can be inserted and the second end **178** of bone preparation guide **150** may define a second fixation aperture through which a second pin can be inserted. These two fixation apertures may be parallel aligned, such that first and second pins **112**, **114** extend through the holes parallel to each other. The first end **176** and/or the second end **178** of bone preparation guide **150** may also define one or more additional fixation apertures that are angled (at a non-zero degree angle) or otherwise skewed relative to the two parallel fixation apertures.

[0062] In use, a clinician may insert the two parallel pins through fixation apertures **180** and may optionally insert one or more angled pins through the one or more angled fixation apertures. This combination of parallel and angled pins may prevent bone preparation guide **150** from being removed from the underlying bones being worked upon. When the clinician has completed using the bone preparation guide, the angled pin or pins may be removed leaving the two parallel pins inserted into the underlying bones. Bone preparation guide **150** can be slid or otherwise moved up and off the parallel pins and, in some examples, a compressor (e.g., a compressor-distractor) thereafter inserted down over the pins. The compressor can then be used to apply a force to the pins to compress the prepared ends of the two facing bones together.

[0063] In some examples as shown in FIG. **8**, bone preparation guide **150** can also include a first adjustable stabilization member **182** engaged with the first end **176** and/or a second adjustable stabilization member **184** engaged with the second end **178**. Each of the members can be threaded and engage a threaded aperture defined by the ends. The elevation of each end can be adjusted with respect to a bone by adjusting the stabilization member. In some embodiments, as shown, the stabilization members are cannulated such that they can receive a fixation pin. In other examples, bone preparation guide **150** does not include stabilization members and/or has a different configuration than that specifically illustrated. Details on example bone realignment instruments and techniques that can be used in conjunction with the present disclosure are described in U.S. Pat. No. 9,622,805, issued Apr. 18, 2017 and entitled "BONE POSITIONING AND PREPARING GUIDE SYSTEMS AND METHODS" the entire contents of which are incorporated herein by reference.

[0064] With reference to FIG. **9**, bone preparation guide **150** may include or be used with a spacer **188** that extends downward from the body **154**. Spacer **188** may be configured to be placed into a joint (e.g., within the TMT joint). In some embodiments, the spacer **188** is selectively engageable with the body of the bone preparation guide and removable

therefrom. The spacer can have a first portion **190** configured to extend into a joint space and a second portion **192** engageable with the body **154**. In the embodiment shown, the spacer can be received within opening **170**, such that the spacer extends from the body in between the first and second guide surfaces. Such a spacer can be useful for positioning the body at a desired position with respect to a joint and for properly positioning the guide with respect to bones to be cut. The distance between the spacer and the first guide surface can define a length of tissue removal (e.g., bone or cartilage to be cut) from a first bone, and the distance between the spacer and the second guide surface can define a length of tissue removal (e.g., bone or cartilage to be cut) from a second bone.

[0065] When configured with spacer **188**, the spacer can be inserted into TMT joint **230** after making incision **400**. After inserting spacer **188** into the TMT joint, opening **170** of bone preparation guide **150** can be aligned with the spacer and the bone preparation guide installed one the spacer, thereby positioning the one or more guide surfaces defined by the bone preparation guide over one or more bone ends to be prepared. In other examples, spacer **188** may be engaged with bone preparation guide **150** prior to installation in TMT joint **230** and the spacer thereafter inserted into the joint as an assembly connected to the bone preparation guide. In still other examples, spacer **188** may be integrally and permanently attached to bone preparation guide **150** or the bone preparation guide may not include a spacer.

[0066] As also shown in FIG. 9, bone preparation guide **150** may include or be used with a tissue removal location check member **194**. Tissue removal check member **194** may be engageable with the body **154** and configured to extend to a first bone and a second bone. The tissue removal location check member can have a first portion **196** configured to extend into contact with first and second bones and a second portion **198** engageable with the body. In the embodiment shown, the tissue removal check member **194** is configured to extend in the body **154** at both the first and second guiding surfaces. The tissue removal location check member **194** may be useful for allowing a practitioner to see where a tissue removing instrument guided by the surfaces will contact the bone to be prepared.

[0067] In addition to or in lieu of using a spacer, the systems and techniques according to the disclosure may utilize a fulcrum. A fulcrum may be a device positionable between the proximal base of the metatarsal being moved (e.g., first metatarsal **210**) and an adjacent metatarsal (e.g., second metatarsal **212**). The fulcrum can establish and/or maintain space between adjacent bones being moved, preventing lateral translation or base shift of the bones during rotation and/or pivoting.

[0068] When used, the fulcrum may be a standalone instrument or may be coupled to spacer **188** to define a bi-planar instrument. A bi-planar instrument can define a spacer body coupled to a fulcrum body that may be useful to provide a unitary structure (e.g., prior to or after being assembled) that can be positioned between two adjacent, intersecting joint spaces: a first joint space between opposed ends of a metatarsal and cuneiform and an intermetatarsal space between adjacent metatarsals. The spacer body can include a first portion insertable into the joint space and a second portion that projects above the joint space. The second portion projecting above the joint space can be coupled to a bone preparation guide, thereby facilitating

positioning of the bone preparation guide over the metatarsal and/or cuneiform between which the spacer body is positioned. The fulcrum body can establish and/or maintain space between adjacent bones being moved, preventing lateral translation or base shift of the bones during rotation and/or pivoting.

[0069] FIG. 10 is a perspective view of an example foot **200** in which a bi-planar instrument is positioned in a first joint space and an intersecting second joint space, where a bone forming the first and second joint spaces is being realigned relative to one or more adjacent bones. In particular, FIG. 10 illustrates a bi-planar instrument having a spacer body **188** coupled to a fulcrum body **189**. Spacer body **188** is positioned at an intersection between an end of first metatarsal **210** and medial cuneiform **222**. Fulcrum body **189** is positioned between first metatarsal **210** and second metatarsal **212**.

[0070] With further reference to FIG. 4, either before or after preparing one or both ends of first metatarsal **210** and medial cuneiform **222**, the clinician may move the metatarsal in at least one plane (**306**). For example, the clinician may move metatarsal **210** in at least the transverse plane to close an intermetatarsal angle between the metatarsal and an adjacent bone (e.g., a second metatarsal) and/or a frontal plane (e.g., to reposition the sesamoid bones substantially centered under the metatarsal). In some examples, the clinician moves the bone portion in multiple planes, such as the transverse plane and/or frontal plane and/or sagittal plane. The clinician may or may not utilize a bone positioning device to facilitate movement of the bone portion.

[0071] FIG. 11 shows a side perspective view of an example bone positioner **10** (also referred to as a bone positioning device) that can be used to move a metatarsal relative to an adjacent bone. In some implementations, the bone positioning device includes a metatarsal engagement member, a tip, and a mechanism to move the metatarsal engagement member and the tip relative to each other in one or more planes. For example, the mechanism may move the metatarsal engagement member and the tip towards each other (e.g. moving the metatarsal engagement member towards the tip, moving the tip towards the metatarsal engagement member, or moving both simultaneously). The bone positioning device may also include an actuator to actuate the mechanism. When the mechanism engaged, it can cause the metatarsal engaged with the metatarsal engagement member to move to correct an alignment in at least one plane with respect to a second bone in contact with the tip.

[0072] In the embodiment of FIG. 11, bone positioning device **10** includes a main body member **20**, a shaft **30**, a metatarsal engagement member **40** is connected to the shaft, and a tip **50** is connected to the main body member. In general, main body member **20** can be sized and shaped to clear anatomy or other instrumentation (e.g., pins and guides) while positioned on a patient. In the embodiment of FIG. 11, the main body member **20** includes a generally C-shaped. In some embodiments, the main body is sized and configured to engage bones of a human foot. In addition, although bone positioning device **10** is illustrated as being formed of two components, main body member **20** and shaft **30**, the guide can be fabricated from more components (e.g., three, four, or more) that are joined together to form the guide.

[0073] A shaft 30 can be movably connected to the main body member 20. In some embodiments, the shaft 30 includes threads 80 that engage with the main body member 20 such that rotation of the shaft translates the shaft with respect to the main body member. In other embodiments, the shaft can slide within the main body member and can be secured thereto at a desired location with a set screw. In yet other embodiments, the shaft can be moved with respect to the main body by a ratchet mechanism or yet other mechanism that rotates and/or linearly translates metatarsal engagement member 40 relative to tip 50. In the embodiment shown, the shaft moves along an axis that intersects the tip 50. In other embodiments, the shaft 30 and/or metatarsal engagement member 40 is offset from tip 50.

[0074] In general, metatarsal engagement member 40 is configured (e.g., sized and/or shaped) to be positioned in contact with an external surface of the skin 402 of the patient. As a result, force applied to and/or through metatarsal engagement member 40 is directed through the skin to the underlying metatarsal 210 rather than directly to the metatarsal. As a result, metatarsal engagement member 40 may be positioned against the skin of the patient without making an incision to facilitate direct contact between the metatarsal engagement member 40 and metatarsal 210.

[0075] FIG. 12 is an illustration showing an example arrangement of bone positioning device 10 in which metatarsal engagement member 40 is positioned in contact with the external surface of skin 402 of the patient covering a side of metatarsal 210. Bone positioning device 10 can apply a force to move metatarsal 210 through skin 402 in one or more planes. In the illustrated configuration, metatarsal engagement member 40 is positioned in contact with the external surface of skin 402 of the patient covering a medial side of metatarsal 210. For example, metatarsal engagement member 40 may be positioned in contact with a portion of skin that covers the medial-most half of metatarsal 210. This arrangement is useful, e.g., to move metatarsal 210 laterally to close the IMA between first metatarsal 210 and second metatarsal 212. In other configurations, the metatarsal engagement member 40 may be positioned in contact with the external surface of skin 402 of the patient covering a different portion of metatarsal 210. In still further implementations, the techniques of the disclosure may be performed without a bone positioning device or a bone positioning device that contacts metatarsal 210 directly rather than through skin 402. However, configuring bone positioning device 10 to deliver movement force through skin 402 can help facilitate a minimal incision procedures.

[0076] With further reference to FIG. 11, metatarsal engagement member 40 may be configured (e.g., sized and/or shaped) to contact skin overlying an underlying bone (e.g., first metatarsal 210). Metatarsal engagement member 40 may define a concave shape to generally conform and/or wrap partially around the underlying cylindrical bone. The concave shape may include define a continuous radius of curvature, a V-shape, a planer region between outwardly extending sidewalls, and/or other shape having a concavity. In still other examples, metatarsal engagement member 40 may be planar. In either case, when configured with a concave shape, the bottom of the concavity can be centered between outwardly extending sidewalls 42, 44. Each sidewall can extend outward to the same height and/or at the same slope, or may extend to different heights and/or at different slopes.

[0077] While the specific dimensions of metatarsal engagement member 40 may vary depending on the manufacturer and/or user preferences, in some examples, the metatarsal engagement member may have a length sufficient at least partially conform around the skin-covered surfaces of the metatarsal being moved (e.g., first metatarsal 210). For example, metatarsal engagement member 40 may have a length 46 extending in a dorsal to plantar direction (when positioned in contact with a patient) that is at least 20 mm, such as at least 25 mm, at least 28 mm, at least 30 mm, or at least 35 mm. For example, the length 46 may range from 28 mm to 35 mm.

[0078] To help position bone positioning device 10 in the sagittal plane when installing on a patient's foot, metatarsal engagement member 40 may include a center line indicator 48. The center line indicator 48 may be a visual indicator (e.g., line) indicating a geometric center of metatarsal engagement member 40 between the dorsal and plantar ends of the metatarsal engagement member. In use, the clinician may substantially align center line indicator 48 with a midpoint (e.g., geometric center between the dorsal and plantar surfaces) of the metatarsal being moved (with the midpoint being covered by skin 402). The clinician may identify the midpoint by visual and/or tactile inspection and/or through radiographic (e.g., fluoroscopic) imaging. Additionally or alternative, the clinician can mark (with a surgical pen) the midpoint on the skin as a reference for alignment. Alignment of the metatarsal engagement member can then occur by aligning the center indicator mark to the marked midpoint of the metatarsal.

[0079] In the embodiment shown, the metatarsal engagement member 40 is provided on an end of the shaft 30. In the embodiment of the shaft shown having threads 80, the metatarsal engagement member 40 can be rotatably coupled to the shaft 30. In such embodiments, as the shaft is rotated relative to the main body member the metatarsal engagement member 40 may or may not rotate with respect to the main body member even as it translates with respect to the main body member along with the shaft 30 and rotates with respect to the shaft. The metatarsal engagement member may oscillate about the shaft 30, but generally does not rotate with respect to skin 402 and/or underlying bone after contact with the skin.

[0080] Tip 50 can be useful for contacting a bone, such as a bone different than the bone being moved by bone positioning device 10. For example, if metatarsal engagement member 40 is in contact with skin covering first metatarsal 210, the tip can be in contact with a lateral side of a different metatarsal (e.g., the second, third, fourth, or fifth metatarsal) and/or a lateral side of skin covering such metatarsal. In some examples, a small stab incision may be made between the different metatarsal and a laterally-adjacent metatarsal. For example, a stab incision may be made in an intermetatarsal space between the second metatarsal 212 and third metatarsal 214. Tip 50 of bone positioning device 10 can be inserted through the incision and the tip positioned in contact with the lateral side of the second metatarsal.

[0081] In different configurations, tip 50 may be straight or may be tapered to facilitate percutaneous insertion and contact with bone. The tip can also include a textured surface, such as serrated, roughened, cross-hatched, knurled, etc., to reduce slippage between the tip and bone. In the embodiment shown, tip 50 further includes a stop 52. Depth stop 52 can limit a depth of insertion into an intermetatarsal

space (e.g., by contacting a dorsal surface of the metatarsal against which tip **50** is intended to be positioned).

[0082] As shown in FIG. **11**, bone positioning device **10** can also include an actuator (e.g., a knob or a handle) **60** to actuate the mechanism, in this embodiment associated with the shaft. In the embodiment shown, the actuator can be useful for allowing a user to rotate the shaft with respect to the main body member **20**. Actuator **60**, shaft **30**, and/or metatarsal engagement member **40** may include a cannulation **70** extending therethrough to allow the placement of a fixation wire (e.g., K-wire) through these components and into contact with or through a bone engaged with the metatarsal engagement member. For example, a fixation wire can be placed into the bone engaged with metatarsal engagement member **40** to fix the position of the metatarsal engagement member with respect to the bone. In another example, the fixation wire can be placed through the bone in contact with the metatarsal engagement member and into an adjacent bone to maintain a bone position of the bone in contact with the metatarsal engagement member and the adjacent bone.

[0083] Embodiments of the bone positioning device may include any suitable materials. In certain embodiments, the bone positioning device is fabricated at least partially from a radiolucent material such that it is relatively penetrable by X-rays and other forms of radiation, such as thermoplastics and carbon-fiber materials. Such materials are useful for not obstructing visualization of bones using an imaging device when the bone positioning device is positioned on bones.

[0084] FIG. **13** is a perspective view of foot **200** showing seeker **188** and fulcrum **189** inserted between adjacent joint spaces and bone positioner **10** engaged with first metatarsal **210**. FIG. **14** is a perspective view of foot **200** from FIG. **13** further illustrated bone preparation guide **150** installed on seeker **188**. In different examples, bone positioner **10** may be installed before and/or after installing bone preparation guide **150** and/or before or after preparing one or both bone ends. FIG. **15** is a further illustration of the surgical instrument arrangement of FIG. **14** showing the example location of incision **400** and skin **402** relative to the surgical instruments.

[0085] Independent of whether the clinician utilizes a bone positioning device or the configuration of such guide, the clinician can move metatarsal **210** in one or more planes. The clinician may move the metatarsal to help correct an anatomical misalignment of the metatarsal. For example, the clinician may move the metatarsal so the metatarsal is anatomically aligned in one or more planes (e.g., two planes, three planes).

[0086] In some examples, the clinician manually moves first metatarsal **210** in addition to or moving the metatarsal through a force applied by bone positioner **10**. For example, the clinician may grasp a pin inserted into the first metatarsal and/or grasp the metatarsal with a tool (e.g., forceps) and manipulate the position of the metatarsal to a desired position. In some applications, bone positioner **10** may apply a force the moves first metatarsal **210** in the transverse plane to close the IMA with an adjacent metatarsal. However, the amount of frontal plane rotation and/or sagittal plane movement achieved by bone positioner **10** through the skin of the patient may be limited. Accordingly, the clinician may manually move first metatarsal **210** (e.g., by grasping a pin inserted into the metatarsal and at least partially projecting

out of the metatarsal) to rotate the metatarsal in the frontal plane and/or move the metatarsal in the sagittal plane.

[0087] In general, an anatomically aligned position means that an angle of a long axis of a first metatarsal relative to a long axis of a second metatarsal is about 10 degrees or less in the transverse plane or sagittal plane. In certain embodiments, anatomical misalignment can be corrected in both the transverse plane and the frontal plane. In the transverse plane, a normal intermetatarsal angle (“IMA”) between a first metatarsal and a second metatarsal is less than about 9 degrees. An IMA of between about 9 degrees and about 13 degrees is considered a mild misalignment of the first metatarsal and the second metatarsal. An IMA of greater than about 16 degrees is considered a severe misalignment of the first metatarsal and the second metatarsal. In some embodiments, the first metatarsal is moved to reduce the IMA from over 10 degrees to about 10 degrees or less (e.g., to an IMA of about 1-5 degrees), including to negative angles of about -5 degrees or until interference with the second metatarsal, by positioning the first metatarsal at a different angle with respect to the second metatarsal.

[0088] With respect to the frontal plane, a normal first metatarsal will be positioned such that its crista prominence is generally perpendicular to the ground and/or its sesamoid bones are generally parallel to the ground and positioned under the metatarsal. This position can be defined as a metatarsal rotation of 0 degrees. In a misaligned first metatarsal, the metatarsal is axially rotated between about 4 degrees to about 30 degrees or more. In some embodiments, methods in accordance with the invention are capable of anatomically aligning the metatarsal by reducing the metatarsal rotation from about 4 degrees or more to less than 4 degrees (e.g., to about 0 to 2 degrees) by rotating the metatarsal with respect to the medial cuneiform.

[0089] In some applications, independent of whether the clinician performs the specific bone realignment technique discussed above, the clinician may compress the end faces of the prepared bones (e.g., first metatarsal **210**, medial cuneiform **222**) together. For example, the technique of FIG. **4** includes optionally compressing the prepared end faces of the bone portions together prior to fixating (**308**). The clinician may compress the end faces together with hand pressure and/or using a compressing instrument physically attached to both the first bone portion and the second bone portion. For instance, the clinician may attach a compressing instrument to metatarsal **210** with one or more fixation pins and also attach the compressing instrument to medial cuneiform **222** using one or more fixation pins. In some examples, the clinician lifts bone preparation guide off two or more pins that extend parallel to each other, at least one of which is inserted into metatarsal **210** and at least one of which is inserted into medial cuneiform **222**. The clinician can then install a compressor back down on the parallel pins.

[0090] When used, the compressing instrument may have a mechanism (e.g., threaded rod, rack and pinion) that presses against the pins inserted through the two bone portions to compress the end faces of the two bone portions together. Additional details on example compressing instruments that may be used can be found in US Patent Publication No. 2020/0015856, published Jan. 16, 2020 and entitled “COMPRESSOR-DISTRACTOR FOR ANGULARLY REALIGNING BONE PORTIONS,” the entire contents of which are incorporated herein by reference.

[0091] FIG. 16 is a perspective view of foot 200 illustrating an example compressing instrument 420 that may be used to compress the prepared end faces of first metatarsal 210 and medial cuneiform 222 together. Compressing instrument 420 may be attached using at least one fixation pin inserted through an arm of the compressing instrument into first metatarsal 210 and using at least one fixation pin inserted through an arm of the compressing instrument into medial cuneiform 222.

[0092] In addition to or in lieu of compressing the prepared end faces of the bones together, the technique of FIG. 4 may involve temporarily or provisionally fixating a moved position of first metatarsal 210 relative to medial cuneiform 220 prior to attaching a permanent fixation device (310). In some implementations, the clinician inserts one or more fixation pins through first metatarsal 210 and into an adjacent bone (e.g., second metatarsal 212, medial cuneiform 222), such as through the end faces of the first metatarsal 210 and medial cuneiform 220 in addition to or in lieu of compressing the end faces with a compressing instrument. The fixation pin may be a k-wire, olive wire (e.g., pin with region of enlarged cross-section) or other fixation pin structure. The fixation pin crossing joint 230 between first metatarsal 210 and medial cuneiform 220 can help provisionally fixate and/or compress the end faces of the two bone portions together prior to installation of permanent fixation device (and subsequent fusion of the bone faces together). When used, the one or more fixation pins can be removed from the end faces of the two bone portions and across the joint between the bone portions after installation of permanent fixation device.

[0093] Independent of whether the clinician provisionally fixates the moved position of first metatarsal 210 relative to medial cuneiform 220, the technique of FIG. 4 involves installing one or more permanent fixation devices to first metatarsal 210 and medial cuneiform 220 across joint 230 (312). The one or more permanent fixation devices may hold the moved position of first metatarsal 210 relative to medial cuneiform 220, allowing the end faces of the bone portions to fuse together through a subsequent healing process.

[0094] Any one or more bone fixation devices can be used including, but not limited to, a bone screw (e.g., a compressing bone screw), a bone plate, a bone staple, an external fixator, a pin (e.g., an intramedullary implant), and/or combinations thereof. The bone fixation device may be secured on one side to metatarsal 210, bridge the TMT joint 230, and be secured on an opposite side to the proximal medial cuneiform 222.

[0095] In one example, two bone plates may be placed across the TMT joint to provide bi-planar plating. For example, a first bone plate may be positioned on a dorsal side of the metatarsal and medial cuneiform. A second bone plate may be positioned on a medial/dorsal-medial side of the metatarsal and medial cuneiform. Independent of the number or configuration of bone plates, the plates may be applied with the insertion of bone screws.

[0096] A variety of different of different bone plate configurations can be used to permanently fixate a metatarsal to a cuneiform. In some examples, the bone plate is configured as a linear bone plate in which all the fixation holes extending through the bone plate are co-linear with each other. In other examples, the bone plate includes one or more fixation holes extending through one or more branch(es) that are non-co-linear with the remaining co-axially arranged

fixation holes. For instance, in various examples, the bone plate may include at least one branch extending outwardly from a longitudinal axis of the bone plate to define at least one of a Y-shape, an L-shape, a T-shape, a U-shape, and/or other shape profile.

[0097] FIG. 17 is illustration of an example bone plate 100 that can be used to help permanently fixate a moved position of a metatarsal relative to an opposed cuneiform. In particular, bone plate 100 illustrates an example bone plate having a U-shaped profile. Configuring bone plate 100 with a U-shaped profile can be useful to help facilitate a minimally invasive surgical procedure. Because of the comparatively short length of the surgical incision in such a procedure, installation of a straight fixation plate through the incision may result in one or more fixation holes being positioned under the skin of the patient and therefore inaccessible for the clinician to insert screws through the fixation holes. By configuring bone plate 100 with a shaped profile (e.g., U-shape), one or more fixation holes extending through the arm(s) of the U-shape may wrap into a region of the incision accessible by the clinician for inserting screws.

[0098] As shown in FIG. 17, bone plate 100 can have a body 102 defining a base 104, a first arm 106 extending at an angle from the base, and a second arm 108 extending at an angle from the base. The first and second arms 106, 108 may extend at the same angle away from base 104 or may extend at different angles. In some examples, the first and second arms 106, 108 each extend at an angle 110 from base 104 ranging from 30 degrees to 175 degrees, such as from 45 degrees to 150 degrees, or from 75 degrees to 135 degrees. For example, angle 110 may be greater than or equal to 90 degrees.

[0099] In either case, bone plate 100 may include one or more fixation holes extending through the thickness of body 102. In these examples, body 102 may include one or more fixation holes in first arm 106, one or more additional fixation holes in second arm 108, and base 104 may or may not include any fixation holes.

[0100] In the illustrated example, first arm 106 has at least one fixation hole, which is illustrated as two fixation holes 112 and 114. Second arm 108 also has at least one fixation hole, which is also illustrated as two fixation holes 116 and 118. Fixation hole 114 is at the intersection of first arm 106 and base 104. Fixation hole 118 is at the intersection of second arm 108 and base 104. Base includes a region between fixation holes 114 and 118 the forms a bridge devoid of any fixation holes, which can be placed bridging across TMT joint 230.

[0101] In the illustrated configuration, one or more drill guides 120 are threadingly inserted into one or more of the fixation holes. Drill guides 120 can provide a structure for guiding a drill to drill into the underlying bone (e.g., prior to inserting a screw). In some implementations, drill guides 120 may also be used to grasp plate 100 with a plate holder to manipulate the plate during placement.

[0102] Although bone plate 100 is illustrated as being configured with four fixation holes the bone plate may include fewer fixation holes or more fixation holes. For example, first arm 106 and second arm 108 of bone plate 100 may each include fewer fixation holes (e.g., one) and/or more fixation holes (e.g., three, four). The dimensions (e.g., length) of the arms and/or base can be adjusted to accommodate the particular number of fixation holes, among other factors.

[0103] Each feature described as a fixation hole may be an opening extending through the thickness of body **102** that is sized to receive a corresponding screw. Each fixation hole may typically have a circular cross-sectional shape although may have other cross-sectional shapes without departing from the scope of the disclosure. The fixation hole may extend perpendicularly through the thickness of body **102** or may taper (e.g., such that the hole is larger adjacent the top surface of the plate than adjacent the bone-facing surface of the plate). In still other examples, the fixation hole may extend at a non-perpendicular angle through the thickness of body **102** and/or be configured for polyaxial screw alignment. In either case, bone plate **100** can be secured to underlying bone portions by inserting fixation elements through corresponding fixation holes. Example fixation elements that can be used are screws, may be locking screws and/or non-locking screws.

[0104] When configured with a U-shape, bone plate **100** may define a maximum length **122** less than the combined length of the individual linear segments of the plate (e.g., the combine lengths **124**, **126**, and **128**). In some examples, maximum length **112** is less than the length of incision **400** whereas the combined length of the individual linear segments of the plate may have a length greater than the length of incision **400**. As a result, fixation holes (e.g., **112**, **116**) that be positioned at a location accessible through incision **400** when bone plate **100** is configured with a U-shape but may otherwise be inaccessible if the same plate is reconfigured as a linear bone plate.

[0105] Access may be particularly challenging when the clinician positions the bone plate at a location offset from the incision location. For example, the clinician may position the base of the U-shaped bone plate at location that is offset at least 45 degrees from the center of the incision, such as at least 60 degrees, at least 75 degrees, or approximately 90 degrees or more. For instance, the clinician may make a dorsal incision and position the base of the U-shaped bone plate on a medial side of the TMT joint. The clinician may position the base of the bone plate across the TMT joint with the legs extending upwardly in a dorsal direction relative to the base.

[0106] FIG. **18** is a perspective view of foot **200** showing an example insertion location of bone plate **100**. As shown in this example, the arms of U-shaped bone plate wrap into region accessible by through incision **400**. In this example, a second bone plate **140** that has a linear profile is attached on a dorsal side of the metatarsal and cuneiform, across the TMT joint.

[0107] To install bone plate **100** (or another bone plate) in a minimal incision procedure, the clinician may utilize a plate holder to help manipulate and/or hold the bone plate within the incision. The plate holder may be an instrument configured to releasably grasp an external surface of the plate and/or be inserted into a fixation hole of the plate, such as a drill guide inserted through the fixation hole. Using the plate holder, the clinician may be able to controllably and precisely position the bone plate and/or hold the bone plate during installation of one or more screws in the small incision created to access the TMT joint space. This can help address space constraints caused by the comparatively small length incision, which can make it difficult for the clinician to properly position the bone plate using their fingers alone.

[0108] In general, a plate holder may be any instrument that can be releasably connected to a bone plate to help

facilitate installation of the bone plate and then detached after and/or during installation. In various examples, the plate holder may be operatively coupled to an exterior surface of the bone plate (e.g., by grasping a protrusion from the plate and/or opposed sides of the plate) and/or an interior surface of the bone plate (e.g., by inserting into one or more fixation holes of the bone plate). Further purposes of installation, when one or more drill guides **120** are inserted into the fixation holes of the bone plate, the drill guides may be considered part of the bone plate that the plate holder can grasp and/or be inserted into.

[0109] FIG. **19** illustrates one example plate holder **450** that is configured as a tenaculum or forceps-style plate holder. In this example, plate holder **450** includes a first arm **452** and a second arm **454** that are pivotably connected together. Each arm extends from a proximal end **456** to a distal end **458**. The distal end of each arm may be linear or, as illustrated, may be curved to define opposed converging radiuses of curvature. When so configured, the arms may be engaged with (e.g., wrapped at least partially about) a feature of bone plate **100**. As illustrated, arms **452**, **454** are wrapped around drill guides **120** inserted into fixation holes of the bone plate. In other examples, the bone plate may include a feature (e.g., a recess, cavity, projection) that one or both arms **452**, **454** can be operatively engaged with (e.g., when the tenaculum is closed) to releasably secure the plate holder to the bone plate. In some examples, plate holder **450** includes a locking mechanism (e.g., ratchet, set screw, etc.) that can lock the arms **452**, **454** at a fixed position (e.g., when grasping plate **100**) but be released by the clinician to remove the plate holder from the plate.

[0110] FIG. **20** illustrates another example plate holder **470** configured to releasably attached to one or more drill guides **120** inserted into one or more fixation holes of a bone plate, such as bone plate **100**. Plate holder **470** may include a handle **472** connected to a body defining one more slots and/or apertures **474** into which a drill guide **120** can be inserted to releasably attach the bone plate to the plate holder.

[0111] FIG. **21** illustrates another example plate holder **480** that is configured as a plate bender-style plate holder. In this example, plate holder **480** defines a shaft **482** terminating in a distal end **484**. In different configurations, the distal end **484** of shaft **482** can be configured (e.g., sized and/or shaped) to be inserted into a fixation hole of a bone plate and/or drill guide **120** inserted into the fixation hole and/or to be inserted over an exterior surface of the drill guide (as illustrated in FIG. **21**). In some examples, plate holder **480** may include a retention mechanism **486**, such as a spring, clasp, O-ring, or other device to provide frictional engagement between the plate holder and bone plate (e.g., drill guide inserted into the bone plate) to help releasably attached the plate holder to the bone plate. In use, the clinician may engage one or more plate holders **480**, such as at least two plate holders positioned on different side of the bone plate bridge that crosses the TMT joint, and use the plate holders to manipulate the position of the plate. Additionally or alternatively, the clinician may use the plate holders to bend the bone plate to a desired shape, e.g., specific to the anatomy of the patient undergoing the procedure. In some examples, incision guide **404** also functions as plate holder **480** and/or a plate bender.

[0112] Independent of the specific configuration of a plate holder may be operatively connected to a bone plate. The

clinician may manipulate the plate holder and, correspondingly, bone plate, to insert the bone plate through incision **400** until the bone plate is positioned across the TMT joint with at least one fixation hole positioned over an underlying metatarsal and at least one fixation hole positioned over an underlying cuneiform. The clinician can then optionally drill a hole into each underlying bone portion through a bone plate fixation hole (e.g., via a drill guide), remove the drill guide, and then insert a screw to secure the bone plate to the underlying bone. In some examples, the clinician installs screws through fixation holes of the bone plate not attached to the plate holder then removes the plate holder and inserts screws through the remaining fixation holes.

[0113] In general, bone plate **100** may be planar (e.g., in the X-Z plane indicated on FIG. **17**) or may include one or more bends or curves that position at least a portion of the bone plate out of plane with a remainder of the bone plate. For example, as illustrated in FIG. **17**, the ends of the legs of the bone plate (e.g., through which fixation hole **112**, **116** are positioned) may be out of plane with the base of the bone plate and/or base **104** may be curved such that fixation holes **114**, **118** are out of plane with each other.

[0114] In some configurations, the curvature of the bone plate may direct drill guides inserted into adjacent pairs of fixation holes to be directed away from one another, e.g., to form a diverging angle relative to each other rather than a converging angle or be parallel to each other. This may be useful to allow the plate holder to be positioned against the outer surface of the diverging drill guide and help prevent the plate holder from inadvertently slipping off the drill guides.

[0115] As an additional feature, the handle and/or arms of the plate holder may be contoured (e.g., curved) out of plane from the proximal end (e.g., where the clinician's hand holds the instrument) to allow for easy placement against a metatarsal. For example, when so configured, the instrument may allow for the bone plate to be placed on the medial side of the metatarsal with the curvature of the instrument generally conforming to the curvature of the anatomy. This conformity can allow for minimal tissue retraction during plate placement.

[0116] FIG. **22** is an illustration of foot **200** showing an example of plate holder **458** manipulating plate **100** for placement across a TMT joint. While example plate holders have generally been described in conjunction with placement of a U-shaped plate, a plate holder may be releasably attached to, and used to position, a plate having a different configuration (e.g., a straight plate) in addition to or in lieu of a U-shaped plate.

[0117] Various examples have been described. These and other examples are within the scope of the following claims.

1. A method of performing a metatarsal correction procedure, the method comprising:

- making an incision to access a tarsometatarsal joint between a first metatarsal and a medial cuneiform;
- using a bone preparation guide to guide a tissue removing instrument to prepare an end of the first metatarsal and an end of the medial cuneiform through the incision;
- using a bone positioning device to apply a force to a skin covering the first metatarsal to move the first metatarsal a transverse plane;
- compressing a prepared end of the first metatarsal and a prepared end of the medial cuneiform together using a compressor; and

applying one or more fixation devices to fixate a moved position of the first metatarsal.

2. The method of claim **1**, further comprising using the bone positioning device engaged with the skin covering the first metatarsal to move the first metatarsal a sagittal plane.

3. The method of claim **1**, further comprising inserting a pin into the first metatarsal and using the pin to move the first metatarsal in a frontal plane.

4. The method of claim **3**, wherein using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse plane and using the pin to move the first metatarsal in the frontal plane comprises correcting a bunion deformity.

5. The method of claim **1**, wherein the bone positioning device comprises a metatarsal engagement member, wherein using the bone positioning device comprises positioning the metatarsal engagement member in contact with the skin covering the first metatarsal.

6. The method of claim **5**, wherein positioning the metatarsal engagement member in contact with the skin covering the first metatarsal comprises positioning the metatarsal engagement member in contact with the skin covering a medial side of the first metatarsal.

7. The method of claim **5**, wherein:

the metatarsal engagement member is on an end of a threaded shaft;

the bone positioning device comprises a knob connected to the threaded shaft; and

using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse plane comprises using the knob to advance the threaded shaft relative to a body of the bone positioning device.

8. The method of claim **7**, wherein a cannulation is defined through the knob, threaded shaft, and metatarsal engagement member, and further comprising inserting a wire through the cannulation into the first metatarsal.

9. The method of claim **1**, wherein the bone preparation guide comprises a slot defined between a first guide surface and a second guide surface, and using the bone preparation guide to guide the tissue removing instrument to prepare the end of the first metatarsal and the end of the medial cuneiform comprises guiding the tissue removing instrument through the slot.

10. The method of claim **9**, wherein the tissue removing instrument comprises a burr.

11. The method of claim **1**, further comprising inserting a spacer into the tarsometatarsal joint and using the spacer to position the bone preparation guide with respect to the tarsometatarsal joint.

12. The method of claim **11**, wherein using the spacer to position the bone preparation guide with respect to the tarsometatarsal joint comprises aligning an opening of the bone preparation guide with the spacer inserted into the tarsometatarsal joint.

13. The method of claim **1**, further comprising, prior to using the bone preparation guide to guide the tissue removing instrument to prepare the end of the first metatarsal and the end of the medial cuneiform through the incision, attaching the bone preparation guide to the medial cuneiform with a pin.

14. The method of claim **1**, wherein using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse

plane comprises using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse plane after using the bone preparation guide to guide the tissue removing instrument to prepare the end of the first metatarsal and the end of the medial cuneiform.

15. The method of claim 1, wherein the bone positioning guide comprises a tip, and further comprising positioning the tip on a lateral side of a second metatarsal separated from the first metatarsal by an intermetatarsal angle.

16. The method of claim 1, wherein making the incision to access the tarsometatarsal joint between the first metatarsal and the medial cuneiform comprises making an incision within a range 3 cm to 4 cm long.

17. The method of claim 1, wherein compressing the prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor comprises compressing the prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor with the compressor attached to the first metatarsal with a first fixation pin and to the medial cuneiform with a second fixation pin.

18. The method of claim 1, wherein compressing the prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor comprises compressing the prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor with the compressor attached to using a same fixation pin inserted into the medial cuneiform used to attach the bone preparation guide to the medial cuneiform.

19. The method of claim 1, wherein applying one or more fixation devices to fixate the moved position of the first metatarsal comprises inserting a wire into the first metatarsal.

20. The method of claim 1, wherein applying one or more fixation devices to fixate the moved position of the first metatarsal comprises applying one or more fixation devices across the tarsometatarsal joint.

21. The method of claim 1, wherein the one or more fixation devices comprise a bone screw and/or an intramedullary implant.

22. A method of performing a metatarsal correction procedure, the method comprising:

making an incision to access a tarsometatarsal joint between a first metatarsal and a medial cuneiform;

using a bone preparation guide to guide a tissue removing instrument to prepare an end of the first metatarsal and an end of the medial cuneiform through the incision, wherein the bone preparation guide comprises a slot defined between a first guide surface and a second guide surface, and using the bone preparation guide to guide the tissue removing instrument to prepare the end of the first metatarsal and the end of the medial cuneiform comprises guiding a burr through the slot;

positioning a metatarsal engagement member of a bone positioning device in contact with a skin covering the first metatarsal and using the bone positioning device to

apply a force to the skin covering the first metatarsal to move the first metatarsal a transverse plane;
inserting a pin into the first metatarsal and using the pin to move the first metatarsal in a frontal plane;
compressing a prepared end of the first metatarsal and a prepared end of the medial cuneiform together using a compressor; and
applying one or more fixation devices to fixate a moved position of the first metatarsal.

23. The method of claim 22, wherein positioning the metatarsal engagement member in contact with the skin covering the first metatarsal comprises positioning the metatarsal engagement member in contact with the skin covering a medial side of the first metatarsal, and further comprising using the bone positioning device engaged with the skin covering the first metatarsal to move the first metatarsal a sagittal plane.

24. The method of claim 22, wherein using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse plane and using the pin to move the first metatarsal in the frontal plane comprises correcting a bunion deformity.

25. The method of claim 22, wherein:

the metatarsal engagement member is on an end of a threaded shaft;

the bone positioning device comprises a knob connected to the threaded shaft;

using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse plane comprises using the knob to advance the threaded shaft relative to a body of the bone positioning device; and

a cannulation is defined through the knob, threaded shaft, and metatarsal engagement member, and further comprising inserting a wire through the cannulation into the first metatarsal.

26. The method of claim 22, further comprising:
inserting a spacer into the tarsometatarsal joint; and
aligning an opening of the bone preparation guide with the spacer inserted into the tarsometatarsal joint to position the bone preparation guide with respect to the tarsometatarsal joint.

27. The method of claim 22, wherein the one or more fixation devices comprise a bone screw and/or an intramedullary implant.

28. The method of claim 22, wherein making the incision to access the tarsometatarsal joint between the first metatarsal and the medial cuneiform comprises making an incision within a range 3 cm to 4 cm long.

29. The method of claim 22, wherein compressing the prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor comprises compressing the prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor with the compressor attached to using a same fixation pin inserted into the medial cuneiform used to attach the bone preparation guide to the medial cuneiform.

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